A Comprehensive Review of Optical Wireless Propagation Models Across Diverse Environments

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***Abstract*—Optical wireless communication (OWC) is gaining prominence as a high-capacity wireless solution, leveraging the unlicensed optical spectrum to support multi-gigabit data rates with energy-efficient deployment. The performance of OWC links, however, is fundamentally shaped by propagation impairments that vary significantly across deployment environments. This survey provides a unified and physics-aware examination of OWC propagation models across atmospheric, indoor, underwater, underground, and space environments, highlighting how each environment introduces distinct attenuation, scattering, turbulence, and geometric effects. To the best of our knowledge, this work offers one of the first comprehensive, cross-environment comparisons of propagation models, together with actionable guidance on model selection, applicability limits, and failure regimes. Models are systematically categorized by environment, spectral regime, and modeling methodology to assess their predictive accuracy and practical relevance. In addition, emerging data-driven and machine learning enhanced approaches are discussed as complementary tools for augmenting physics-based models in complex or nonstationary conditions. The survey aims to provide researchers and practitioners with a structured reference for selecting, evaluating, and developing propagation models that enable robust and scalable optical wireless systems for next-generation communication networks.**

***Index Terms*—Free-space optical communication, propagation modeling, atmospheric turbulence, underwater optical wireless communication, optical wireless networks, machine learning based channel modeling.**

List of Abbreviations

AWGN	Additive White Gaussian Noise
BER	Bit Error Rate
CNN	Convolutional Neural Network
CSI	Channel State Information
DC	Direct Current
DNN	Deep Neural Network
DSL	Deep-Space Link
DSOC	Deep-Space Optical Communication
ET	Edge Transmitter Set
ESA	European Space Agency
FDTD	Finite-Difference Time-Domain
FET	Field-effect Transistor
FoV	Field-of-View
FSO	Free-Space Optical
GBR	Gradient Boosting Regressor
G2S	Ground-to-Space
HAP	High-Altitude Platform
HPE	High Photon Efficiency
IM/DD	Intensity Modulation / Direct Detection
IoT	Internet-of-Things
IoUT	Internet of Underground Things
ISI	Inter-Symbol Interference
ISL	Inter-Satellite Link
KNN	k-Nearest Neighbors
LoS	Line-of-Sight
ML	Machine Learning
MLP	Multi-Layer Perceptron
MIMO	Multiple-Input Multiple-Output
NARX	Nonlinear Autoregressive with Exogenous Inputs
NIR	Near-Infrared
NLoS	Non-Line-of-Sight
NMSE	Normalized Mean Square Error
OOK	On-Off Keying
OSNR	Optical Signal-to-Noise Ratio
OWC	Optical Wireless Communication
PAT	Pointing, Acquisition, and Tracking
PD	Photodiode
PDF	Probability Density Function
PPM	Pulse Position Modulation
PR	Received Optical Power
PT	Transmitted Optical Power
QoS	Quality of Service
RF	Radio Frequency
RL	Reinforcement Learning
RMSE	Root Mean Square Error
RSSI	Received Signal Strength Indicator
SARSA	State–Action–Reward–State–Action
SINR	Signal-to-Interference-plus-Noise Ratio
6G	Sixth-Generation
SNR	Signal-to-Noise Ratio
SVR	Support Vector Regression
SVM	Support Vector Machine
SWIR	Short-Wave Infrared
SWaP	Size, Weight, and Power
UAV	Unmanned Aerial Vehicle
UWC	Underwater wireless communication
UOWC	Underwater Optical Wireless Communication
UWSN	Underwater Wireless Sensor Network
VLC	Visible Light Communication
WDM	Wavelength Division Multiplexing
WIRC	Wireless Infrared Communication
XGBoost	Extreme Gradient Boosting

I. Introduction

In the current landscape of wireless communications, radio frequency (RF) and microwave technologies remain the backbone of global connectivity. However, escalating spectrum scarcity, high licensing costs, and the densification of

next-generation networks have pushed conventional RF-based systems to their limits. As new paradigms such as sixth-generation (6G) networks, the Internet-of-Things (IoT), and ultra-reliable low-latency communications emerge, there is a growing need for complementary solutions that can alleviate the burden on RF spectrum while delivering high capacity, energy efficiency, and scalability. Optical wireless communication (OWC)-often used interchangeably with free-space optical (FSO) communication-has gained prominence as a promising candidate to meet these requirements [1], [2]. By exploiting the vast, unlicensed optical spectrum, OWC systems can provide fiber-like data rates, immunity to electromagnetic interference, enhanced security, and rapid deployability. Rather than replacing RF, OWC is increasingly viewed as a complementary technology that can offload traffic, extend coverage, and enable hybrid RF/optical infrastructures for resilient next-generation networks [3].

The fundamental principle of OWC involves transmitting modulated optical carriers—typically in the visible, near-infrared, or mid-infrared bands—between a transmitter and receiver through an unguided medium such as water or open space. Despite its attractive features, OWC link performance is strongly dependent on the propagation medium. Environmental factors such as atmospheric turbulence, absorption and scattering by fog, haze, and rain, underwater turbulence and particulate absorption, dust and humidity, or multipath interference in indoor environments impose significant impairments on the optical wireless channel [4]. As a result, specialized propagation models have been developed to characterize and mitigate these impairments across different deployment scenarios. For instance, atmospheric models capture turbulence-induced scintillation and weather-dependent attenuation, underwater models emphasize optical scattering and absorption, while space models address solar background radiation and vacuum propagation. Underground environments must contend with obstacles, soil composition, and humidity, while indoor optical channels face multipath and shadowing.

A large body of work has addressed these propagation challenges using analytical, empirical, and hybrid models. Classical empirical attenuation models such as those by Kaushal *et al.* [4], Kruse *et al.* [5], and Al Naboulsi *et al.* [6] remain widely used for fog and visibility-dependent attenuation, while more recent studies emphasize site-specific measurements and region-dependent prediction methods [1]. At the same time, statistical turbulence models such as log-normal, gamma–gamma, and Málaga distributions have been developed to capture scintillation effects across weak-to-strong turbulence regimes [4]. Wavelength dependence is another critical factor: widely adopted operating bands include 850 nm, 1064 nm, and 1550 nm, each offering different trade-offs between attenuation, scattering sensitivity, and eye safety [1]. Recent surveys also highlight the role of multi-beam OWC systems, hybrid RF/optical backhauls, and emerging data-driven modeling approaches as means to improve reliability and scalability in harsh environments [1], [3]. Collectively, these developments underline the importance of a unified understanding of propagation models across all environments—aerial, indoor, underwater, underground, and space—as a prerequisite for deploying resilient OWC systems in next-generation wireless infrastructures.

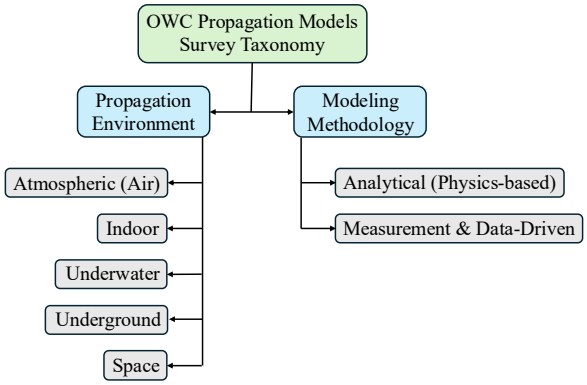


Fig. 1: Two-level classification of OWC channel models by environment and modeling approach.

This survey aims to establish a unified framework for OWC channel modeling by systematically classifying and evaluating existing propagation approaches. We introduce a two-level taxonomy as shown in Fig. 1, first based on the physical environment and then on the modeling methodology, to organize the diverse literature into a coherent structure. By presenting models across the full optical spectrum and all deployment scenarios, we offer a structured overview of techniques suited to different environmental conditions and wavelength regimes. Our scope extends from classical analytical formulations to cutting-edge, data-driven, and machine learning (ML)-augmented methods, highlighting their relative merits and limitations. In so doing, we illuminate key challenges in OWC channel characterization such as turbulence, scattering, beam wander, and background noise, and identify emerging trends and research opportunities that will inform the design of robust, high-performance OWC systems for real-world applications. To further bridge the gap between theory and practice, we provide an open-source Python library that implements the core models discussed in this survey, offering a ready-to-use tool for the OWC research community.

Key contributions of this survey are as follows:

1) **A Two-Level Taxonomy:** We introduce a structured, physics-aware classification of OWC propagation models based on:
   - *Level 1:* physical environment (atmospheric, indoor, underwater, underground, space).
   - *Level 2:* modeling methodology (analytical (physics-based), measurement and data-driven).
2) **Unify Cross-Environment Frameworks:** Consolidate propagation models across all major optical wavelengths into a single comparative framework, highlighting common assumptions, modeling parameters, and environmental differences.
3) **Evaluate Model Applicability and Limits:** Categorize models by environmental factors and spectral regimes to discuss their specific failure regimes, applicability limits, and suitability for next-generation deployment scenarios.

4) **Review of ML Integration:** Review the integration of ML into OWC propagation modeling, including current applications and prospective future directions.
5) **Open-Source Implementation:** Develop and release an open-source Python code-base implementing representative OWC propagation models discussed in this survey, enabling reproducibility, benchmarking, and direct adoption by the research community.

The rest of the survey is organized as follows: Section II reviews existing literature surveys across different propagation environments and modeling methodologies, highlighting the scope and limitations of prior surveys. Section III introduces a data-driven propagation modeling framework, presenting empirical and ML-based approaches and clarifying their methodological foundations and practical roles. Section IV examines atmospheric (air) propagation models by integrating analytical formulations with empirical and ML-based methods under a unified perspective. The subsequent sections extend the discussion to indoor, underwater, underground, and space environments, emphasizing their distinct physical impairments and modeling challenges. Finally, the survey concludes with a critical discussion of current modeling challenges and provides recommendations and future research directions toward unified, scalable, and cross-environment OWC propagation frameworks.

II. Related Work

Existing surveys on OWC propagation models are largely organized around specific deployment environments or modeling methods, in some cases, around a spectral regime as shown in Fig. 2. Prior work has provided in-depth coverage of FSO links, indoor visible light communication (VLC) channels, underwater OWC (UOWC) systems, underground environments, and deep-space optical links, often focusing on the dominant physical impairments and mitigation techniques within each setting. While these studies have significantly advanced the understanding of environment-specific propagation phenomena, they are typically developed in isolation and do not offer a unified perspective on how modeling assumptions, parameter choices, and limitations differ across environments. To highlight this fragmentation and to motivate the need for a cross-environment synthesis, we review the existing literature below by grouping prior surveys according to their primary deployment environments.

A. Atmospheric (Air)

Khalighi *et al.* [7] provide a comprehensive overview of FSO communication systems from a communication-theoretic perspective. The first part of the study focuses on FSO link characterization, detailing channel models and transmitter/receiver structures. The latter part investigates the information-theoretic limits of FSO channels and reviews algorithmic-level system designs aimed at approaching these limits, including modulation, channel coding, spatial and cooperative diversity techniques, adaptive transmission, and hybrid RF/FSO systems.

Al-Gailani *et al.* [1] reviews recent advances in FSO technology up to 2020 and discusses the factors driving its evolution toward ubiquitous deployment. It presents fundamental FSO system concepts, including single-beam and multi-beam architectures, examines the effects of rain and haze on signal propagation, and addresses network scalability through hybrid multi-beam FSO systems employing wavelength-division multiplexing (WDM).

More recently, Sahu *et al.* [8] examines FSO communication as a key enabler for 5G and beyond-5G networks, highlighting both its advantages and deployment challenges stemming from atmospheric impairments, system complexity, and link reliability. The survey covers advanced FSO concepts such as artificial intelligence (AI)/ML-assisted adaptive control, Network Functions Virtualization or Software-defined Networking based resource management, Unmanned Aerial Vehicle (UAV) or High-Altitude Platform (HAP)-assisted links, energy-efficient and quantum FSO architectures, reconfigurable intelligent surface-based optimization, and comparative channel modeling under varying turbulence conditions. Sarkar *et al.* [9] reviews available propagation models for outdoor (Okumura *et al.* Model, Hata Model, COST-231-Walfisch-Ikegami Model, Dual-Slope Model) environments, classifying them into statistical and site-specific approaches, also known as deterministic models, while discussing key radio channel characteristics such as path loss, fading, and time-delay spread. Ghassemlooy *et al.* [10] addresses the dynamic characteristics of the environment and presents models that characterize the statistical behavior of the atmospheric channel, focusing primarily on fog and turbulence.

B. Indoor

Al-Kinani *et al.* [11] highlights the applicability of Wireless Infrared Communication (WIRC) and VLC technologies in indoor settings for different link configurations i.e., directed line-of-sight (LoS), non-directed LoS, diffused, and tracked links. Miramirkhani *et al.* [12] discusses analytical and simulation-based indoor channel models for VLC, also provides a comparison of existing modeling techniques related to mobile VLC channel. Sarker *et al.* [9] discusses foundational indoor propagation models including distance-power model, Finite-Difference Time-Domain (FDTD) model, ray-tracing, and hybrid models, while classifying them into statistical and deterministic modeling approaches. Along with atmospheric models, Ghassemlooy et al. [10] also contribtue to indoor VLC channel characteristics, providing deeper insight on optical channel behavior in dynamic indoor environments.

C. Underwater

Anguita *et al.* [13] provide an overview of OWC in the context of underwater optical sensor networks, outlining both its potential advantages and the key challenges associated with its use. While Moffione *et al.* [14] focuses on methodologies for measuring optical properties in marine environments, it does not equip readers with the essential channel models of UOWC. Johnson *et al.* [15] examine the physical mechanisms

that govern how light behaves underwater—such as absorption, scattering, and refractive-index variations—and clearly distinguishes between inherent and apparent optical properties used to differentiate various water types. However, the work does not provide a communication-oriented UOWC channel model; rather, it focuses on explaining the fundamental ocean-optics principles that influence light propagation. Sajmath *et al.* [16] present a comprehensive survey of existing studies, highlighting the challenges and technologies currently employed in UOWC, and discuss various potential applications. Arnon [17] models three types of UOWC: LoS link, modulating retroreflector link and reflective link. Kaushal *et al.* [18] include these models in their survey to provide an overview of UOWC channels and associated research challenges, summarize UOWC systems demonstrated up to 2015 with detailed system parameters, and present an exhaustive review of communication challenges.

D. Underground

Subterranean environments encompass a wide range of confined and enclosed settings, including underground mining operations, road and rail tunnels, metro/subway systems, pipelines, utility and service galleries, caves, and underground shelters. These environments are characterized by the absence of natural light, highly irregular geometries, and the presence of particulate matter such as dust, moisture, and gas, all of which significantly affect optical wave propagation.

Unlike atmospheric and UOWC, underground optical propagation has received relatively limited attention. Existing studies are primarily focused on specific scenarios such as underground mining environments and tunnel-like structures, where LoS optical links can be established over short distances. Saeed *et al.* [19] provide a comprehensive survey of communication technologies for the Internet of Underground Things (IoUT), including electromagnetic, acoustic, magnetic induction, and wired methods. While the survey briefly mentions VLC as a potential technique, it does not provide detailed modeling or analysis of optical wireless propagation in underground environments. Riurean *et al.* [20] present a detailed propagation analysis of VLC in underground mining environments, examining the effects of dust-induced scattering and reflective tunnel surfaces on optical signal propagation. Their results show that these environmental factors significantly influence channel characteristics and overall communication performance.

E. Deep Space

Karmous *et al.* [21] provide a comprehensive survey of deep-space optical communication technologies, emphasizing emerging methods to enhance link capacity, reliability, and security for future missions. Wu *et al.* [22] review the evolution, principles, and core technologies of deep-space optical communication systems while identifying development trends for next-generation laser links. Ivanov *et al.* [23] examine European Space Agency (ESA)-supported deep-space laser communication programs, presenting next-generation high photon efficiency (HPE) link architectures and prospects for missions extending beyond 4.2 AU. Kaushal and Kaddoum [24] survey major challenges in atmospheric and space-based FSO links and discuss mitigation strategies, including turbulence compensation and orbital angular momentum techniques, to improve link performance. Li *et al.* [25] provide an extensive overview of laser space network terminals, inter-satellite links, and architectural designs, outlining current progress and future directions for building secure, high-speed space communication networks.

The scope and core contributions of the related work discussed in this section are concisely summarized in Table I. In this survey, we make a unifying review of the FSO propagation models and bring together the prominent models for each environment as a one-source reference. The coverage of existing surveys are also depicted in Table II which shows the comprehensiveness of this work. We also develop a Python code-base for these models, making them freely accessible to researchers.

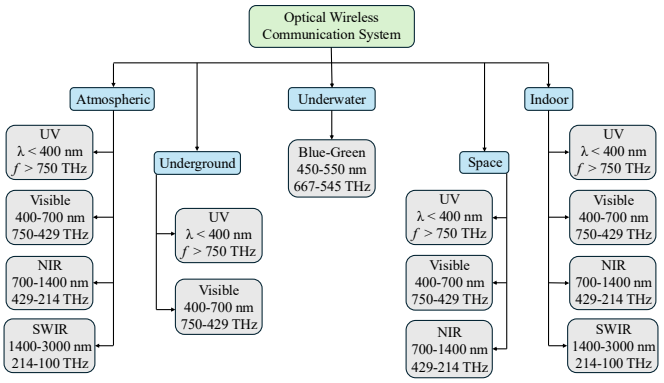


Fig. 2: Classification of OWC systems by environment and spectral regime.

III. Data-Driven Propagation Modeling Framework

As motivated in the related work discussion, data-driven approaches play an increasingly important role in OWC channel characterization. Unlike purely analytical formulations derived from first principles, data-driven models rely on measurement-based calibration or learned mappings to capture complex environmental interactions. In this section, we first describe empirical modeling approaches, followed by ML-based propagation models, and conclude with a structured comparison between these two paradigms.

A. Empirical Modeling in OWC Channel Characterization

An *empirical model* is a data-driven representation of system behavior derived from experimental observations rather than purely theoretical first principles. In OWC, these models utilize field measurements—such as meteorological visibility (V), received optical power (P_{R} in W), and atmospheric attenuation—to capture complex environmental interactions often simplified in analytical derivations. These models are essential for capturing practical impairments, hardware-specific limitations, and local environmental variations.

TABLE I: Summary of related work categorized by environment and research focus.

Environment	Paper	Year	Focus / Contribution
Air	[7]	2014	Reviews FSO channel models, transceiver designs, atmospheric impairment mitigation strategies with algorithmic and hybrid RF/FSO solutions.
	[1]	2020	Reviews FSO system architectures, propagation impairments, scalability, and multi-beam WDM solutions.
	[8]	2025	Surveys advanced FSO technologies for 5G/B5G, covering atmospheric impairments, AI/ML-enabled hybrid architectures, UAV/HAP-assisted links, and system-level performance trade-offs.
Indoor & Air	[9]	2003	Reviews radio propagation models for indoor (distance-power, FDTD, ray tracing, hybrid) and outdoor (Okumura, Hata, COST-231, Walfisch–Ikegami, dual-slope) environments.
	[10]	2019	Surveys OWC channel models; addresses dynamic atmospheric characteristics, fog and turbulence modeling, and provides partial coverage of indoor VLC channel models.
Indoor	[11]	2018	Surveys WIRC and VLC indoors; compares link configurations: directed LoS, non-directed LoS, diffused links, and tracked links.
	[12]	2020	Reviews analytical and simulation-based VLC indoor channel models; compares modelling approaches for mobility-aware VLC channels.
Underwater	[14]	2001	Focuses on experimental characterization of inherent optical properties of seawater, providing measurement methodologies but without formulating communication-oriented UWOC channel models.
	[17]	2010	Models three underwater optical communication link types: LoS link, modulating retro-reflector link, and reflective link.
	[13]	2010	Discusses the prospects of OWC in UWSN, emphasizing system-level benefits, deployment challenges, and the role of optical links in dense sensing environments.
	[15]	2012	Explores key drivers of underwater light propagation, including absorption, scattering, and refractive-index effects, and distinguishes between inherent and apparent optical properties across water types.
	[18]	2016	Surveys underwater optical channels, research challenges, system demonstrations up to 2015, and key parameters affecting UOWC.
	[16]	2020	Comprehensive survey of UWOC systems; summarizes challenges, technologies, and application domains.
Underground	[19]	2019	Surveys communication technologies for IoUT, including electromagnetic, acoustic, magnetic induction, and visible light methods, with emphasis on underground propagation challenges and system design.
	[20]	2021	Analyzes VLC propagation in underground mining environments, focusing on dust-induced scattering and reflections from tunnel surfaces and their impact on channel characteristics.
Space	[24]	2016	Surveys atmospheric and space-based FSO impairments and mitigation, covering turbulence effects, link-layer solutions, and methods like orbital angular momentum to improve capacity and reliability.
	[22]	2018	Reviews the evolution, principles, and key technologies of optical communication in deep space, highlighting its ability to meet rising data demands and outlining future directions for deep-space laser communication development.
	[25]	2022	Reviews the terminals, links, and architectures for laser space networks, outlining future directions for building high-speed, secure, next-generation space communication systems.
	[21]	2024	Provides a comprehensive overview of deep space technologies and emerging methods to enhance capacity, reliability, and security for future deep-space missions.
	[23]	2025	Reviews ESA-supported deep-space laser communication developments, examines next-generation HPE optical link architectures, and outlines the technological prospects for future long-distance missions beyond 4.2 AU.

Empirical modeling provides several key benefits for the design and analysis of OWC links:

- Deployment Guidance: Empirical findings offer reliable benchmarks for link designers, aiding in the planning of permanent OWC installations and reducing the risk of outages in specific geographical regions.
- Performance Optimization: Measured data enables the refinement of system parameters, such as transmitted power, wavelength selection, and aperture size, to maximize link reliability and energy efficiency.
- Validation of Analytical Models: Empirical models provide reference measurement data for validating analytical and simulation-based OWC models, ensuring that theoretical predictions align with observed performance.
- Standardization and Planning: Insights derived from empirical measurements contribute to defining realistic performance targets and availability standards for next-generation optical wireless networks.

1) Common Structure of Empirical OWC Models: Beyond their conceptual advantages, empirical OWC models typically follow a structured development process consisting of five primary stages:

- Data Collection: Channel measurements are obtained through outdoor field trials or controlled laboratory experiments. Typical datasets include P_{R}, V, temperature (T), humidity, and link-specific parameters such as range and wavelength.
- Preprocessing and Metric Extraction: Raw measurements are processed to compute key metrics, such as the scintillation index:

$$\sigma_P^2 = \frac{\mathbb{E}[P_{\mathrm{R}}^2]}{\mathbb{E}[P_{\mathrm{R}}]^2} - 1, \tag{1}$$

or normalized attenuation levels. $\mathbb{E}[\cdot]$ denotes the expectation operator. Preprocessing includes outlier removal and

TABLE II: Coverage comparison of existing surveys across propagation environments.

Ref.	Year	Air	Indoor	Underwater	Underground	Space
[7]	2014	✓	✗	✗	✗	✗
[1]	2020	✓	✗	✗	✗	✗
[8]	2025	✓	✗	✗	✗	✗
[9]	2003	✓	✓	✗	✗	✗
[10]	2019	✓	✓	✗	✗	✗
[11]	2018	✗	✓	✗	✗	✗
[12]	2020	✗	✓	✗	✗	✗
[14]	2001	✗	✗	✓	✗	✗
[17]	2010	✗	✗	✓	✗	✗
[13]	2010	✗	✗	✓	✗	✗
[15]	2012	✗	✗	✓	✗	✗
[18]	2016	✗	✗	✓	✗	✗
[16]	2020	✗	✗	✓	✗	✗
[19]	2019	✗	✗	✗	✓	✗
[20]	2021	✗	✗	✗	✓	✗
[24]	2016	✓	✗	✗	✗	✓
[22]	2018	✗	✗	✗	✗	✓
[25]	2022	✗	✗	✗	✗	✓
[21]	2024	✗	✗	✗	✗	✓
[23]	2025	✗	✗	✗	✗	✓
This Work	2026	✓	✓	✓	✓	✓

segmentation based on weather conditions, e.g., fog, rain, or snow.

- Model Fitting: Extracted metrics are mapped to deterministic attenuation laws or statistical fading distributions. For visibility-based models, the mapping typically follows:

$$A_{\text{fog}} = \frac{\mathcal{J}}{V}\left(\frac{\lambda}{\lambda_0}\right)^{-\mathcal{Q}}, \tag{2}$$

where $\mathcal{J}$ and $\mathcal{Q}$ are empirically fitted coefficients, λ is the operating wavelength, and λ_0 is the reference wavelength (typically $550\,\text{nm}$). Alternatively, for fading, distributions such as the log-normal probability density function (PDF) are utilized:

$$f_{\text{LN}}(P_{\text{R}}) = \frac{1}{P_{\text{R}}\sigma\sqrt{2\pi}}\exp\left[-\frac{(\ln P_{\text{R}} - \mu)^2}{2\sigma^2}\right], \tag{3}$$

where μ and σ denote the mean and standard deviation of the logarithmic received power, capturing the average channel attenuation and turbulence-induced fluctuations, respectively.

- Validation and Error Analysis: Models are validated using reserved data subsets or comparative evaluation against established baselines (e.g., Kim *et al.* [26] or Kruse *et al.* [5] models). Standard metrics include Root Mean Square Error (RMSE) and the coefficient of determination (R^2).
- Application: Once validated, models are used for predicting link availability, generating simulation channel traces, or integrating into higher-layer metrics such as ergodic capacity (C) and bit error rate (BER).

B. ML–Based Propagation Models

Complementary to parametric empirical approaches, ML-based propagation models adopt a data-driven learning paradigm. ML approaches for propagation modeling are increasingly explored in OWC systems because they can learn complex, nonlinear, and environment-dependent relationships directly from data. Unlike analytical models, which require explicit physical parameterization (e.g., scattering/absorption coefficients, turbulence parameters, and pointing-error statistics), and unlike many empirical models, which often assume a predefined functional form (e.g., visibility-based attenuation laws), ML models can integrate multiple heterogeneous inputs (meteorological sensors, received-signal features, platform-motion data) to predict link metrics such as received power, turbulence strength, outage/availability, or detection performance [27]. In this sense, ML-based modeling is best viewed as a complementary layer that either (i) directly maps observable features to link metrics, or (ii) augments physics-based models through data-driven parameter estimation and fast surrogate prediction.

1) Common Structure of ML-Based Propagation Modeling: Most ML-based propagation modeling studies follow a shared pipeline:

- Data collection and labeling: Measurements from field trials or controlled experiments (or physics-based simulators) are used to form input features (e.g., T, relative humidity (RH), wind speed, solar flux, received signal strength indicator (RSSI) samples, histogram features) and output labels (e.g., P_{R}, Optical Signal-to-Noise Ratio (OSNR), $\log_{10} C_n^2$ (where C_n^2 denotes the refractive index structure parameter), outage probability, BER, or availability class).
- Preprocessing and feature engineering: Datasets are synchronized, cleaned, and transformed into learning-ready representations (e.g., scalar weather vectors, waveform statistics, or histogram/image-like features). When derived quantities are used (e.g., C_n^2), they are obtained from the measurement campaign or from an accompanying physical model.
- Model training: A supervised learning model is trained to learn a mapping $\hat{y} = f(\mathbf{x})$, where $\mathbf{x}$ denotes the feature vector and y the target metric. Common choices include tree-based regressors such as random forest and gradient boosting regression (GBR), kernel-based methods such as support vector regression (SVR), nearest-neighbor approaches such as k-nearest neighbors (KNN), and neural network architectures including multilayer perceptron (MLP), convolutional neural network (CNN), and deep neural network (DNN), depending on whether inputs are tabular features or structured representations [28], [29].
- Evaluation and validation: Models are evaluated on held-out data using appropriate metrics (e.g., RMSE and R^2 for regression; accuracy/precision/recall/F1 for classification; BER/Q-factor for detection tasks). Cross-validation and careful train/test splits are used to limit overfitting and quantify generalization [30], [31].
- Deployment and integration: Trained models are integrated into monitoring or control loops to enable rapid prediction of link quality and to support adaptation (e.g., proactive power/beamwidth control, availability forecast-

TABLE III: Comparison between empirical and ML-based propagation modeling.

Feature	**Empirical Modeling**	**ML-Based Modeling**
Functional Structure	Predefined analytical form with fitted coefficients (e.g., power law, log-normal PDF).	Learned functional mapping without imposing a fixed analytical equation.
Model Capacity	Low-dimensional parameterization (typically 1–3 fitted parameters).	High-dimensional parameterization with potentially large numbers of learned weights.
Input Dimensionality	Limited inputs (e.g., visibility V, wavelength λ, link range).	Multi-feature inputs (e.g., temperature, humidity, wind speed, solar flux, received-signal statistics).
Interpretability	High interpretability; parameters often have physical meaning.	Reduced interpretability; internal representations may not directly correspond to physical quantities.
Data Requirement	Can be calibrated with relatively modest measurement datasets.	Typically requires larger datasets to ensure generalization and avoid overfitting.
Computational Cost	Minimal evaluation cost; simple closed-form expressions.	Low inference cost after training; training and hyperparameter tuning may be computationally intensive.
Handling of Nonlinearity	Limited to the nonlinear structure of the chosen analytical form.	Capable of modeling complex, coupled, and high-order nonlinear relationships.
Generalization Behavior	Broad but coarse-grained; may not fully capture region-specific aerosol or turbulence variability.	Often site-dependent; may require retraining, fine-tuning, or domain adaptation for new environments.
Primary Deployment Role	Long-term link budgeting, availability estimation, and standards development.	Real-time monitoring, adaptive control (e.g., Pointing, Acquisition, and Tracking (PAT)), outage prediction, and impairment estimation.
Typical Estimation Method	Linear or nonlinear least-squares curve fitting.	Gradient-based optimization, ensemble learning, or reinforcement learning (RL).

ing, or sensorless parameter monitoring).

Empirical vs. ML-based propagation modeling. Empirical and ML-based propagation models are both data-driven, but they differ in *how* data are used and *what* the model provides, as shown in Table III. Empirical FSO models typically calibrate a physically interpretable, low-dimensional functional form (e.g., visibility-based attenuation or parametric fading PDFs), which makes them easy to validate, explain, and integrate into link-budget tools, but can limit accuracy when multiple correlated factors jointly affect the channel. In contrast, ML-based models learn a direct mapping from measured features (meteorology, received-signal statistics, or platform states) to link metrics (e.g., received power, outage/availability, C_n^2, or detection outcomes), enabling flexible nonlinear modeling and fast inference once trained. This flexibility comes at the cost of stronger dependence on data representativeness and careful generalization testing. Consequently, empirical models remain valuable as interpretable baselines and deployment tools, while ML models are especially attractive for real-time monitoring and adaptation in scenarios where multi-parameter interactions are difficult to capture with fixed-form empirical expressions.

The following sections will describe the propagation challenges, analytical and empirical models, and recent ML-based approaches for each of these key OWC environments, providing a structured overview of OWC channel characterization across diverse deployment scenarios. For better accessibility, description of symbols used in different optical wireless propagation environments are summarized in Table IV.

IV. Atmospheric (Air) Propagation Models

The atmospheric environment, a.k.a. air, encompasses the region within Earth's atmosphere. The atmospheric channel is a complex and dynamic environment where the characteristics of the propagating optical beam is influenced by weather phenomena, particulates, and turbulence.

A. Propagation Challenges

Fig. 3 summarizes the major physical impairments that affect optical signal propagation in atmospheric (air) environments. The most prominent of these challenges are as follows:

- Atmospheric Turbulence: Causes scintillation and signal fading due to temperature and pressure variations.
- Weather Conditions: Fog, rain, snow, and other adverse weather conditions significantly attenuate optical signals.
- Aerosols and Particulates: Absorption and scattering caused by water vapor and aerosols reduce signal intensity. The attenuation of the optical signal in the atmosphere is due to scattering and absorption introduced by the molecular constituents (gases) and aerosols [10].
- Multipath Interference: Reflections from the ground or nearby objects can cause multipath fading.

B. Propagation Model

This section develops the signal-to-interference-plus-noise ratio (SINR) based propagation model used for evaluating OWC performance under geometric spreading, atmospheric attenuation, atmospheric turbulence, and stochastic pointing errors induced by (UAV or mobile) platform vibration. The key geometric components of the model are illustrated in Fig. 4. Rather than relying on a single closed-form range equation, we adopt the widely used multiplicative channel decomposition:

$$P_{\mathrm{R}} = P_T \; h_{\mathrm{geo}} \; h_{\mathrm{atm}} \; h_{\mathrm{turb}} \; h_{\mathrm{pe}}, \tag{4}$$

where P_T (W) is the transmitted optical power. The term h_{geo} captures geometric attenuation, h_{atm} represents the atmospheric transmittance based on the Beer–Lambert law, h_{turb} models scintillation due to turbulence, and h_{pe} denotes

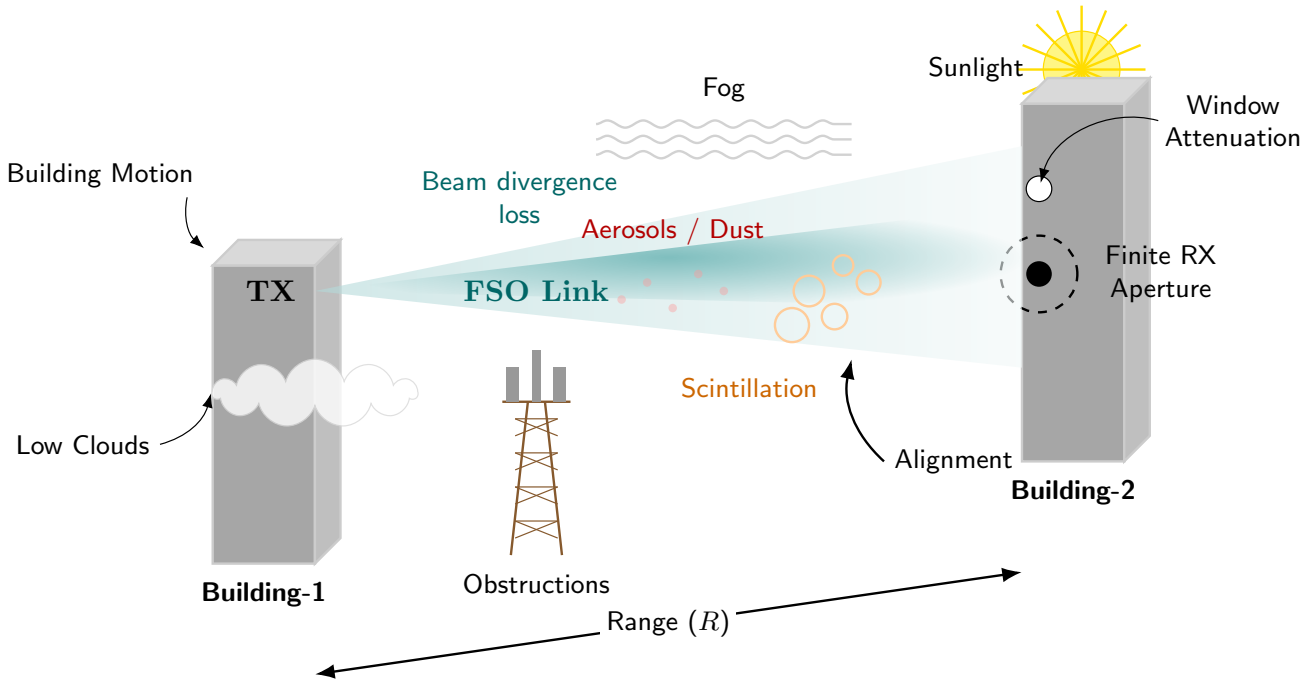


Fig. 3: Impairments affecting atmospheric FSO communication links.

stochastic pointing-error loss arising from platform vibration and tracking instability. This formulation provides a unified basis for incorporating both classical models and the data-driven UAV vibration models considered later.

1) Fundamental FSO Range Equation: Although (4) provides a modular representation of the OWC channel, classical analyses frequently begin with the FSO range equation, which combines geometric spreading, optical coupling, and atmospheric attenuation [32]. The received power is expressed as [33]:

$$P_{\mathrm{R}} = P_T \left[\frac{A_{\mathrm{R}}}{(D_T + R\,\theta)^2} \right] \mathcal{T}\, \mathcal{K}\, \exp\!\left(-\alpha_{atm} R\right), \tag{5}$$

where A_{R} is the receiver aperture area (m^2), D_T is the transmitter diameter (m), θ is the full divergence angle (rad), α_{atm} is the atmospheric attenuation coefficient (m^{-1}), R is the link range (m), and the bracketed term denotes geometric loss. The factors $\mathcal{T}$ and $\mathcal{K}$ account for internal optical efficiencies and source coupling, while the term $\exp\!\left(-\alpha_{atm} R\right)$ captures atmospheric attenuation.

The subsequent subsections reinterpret these components within the generalized framework of (4).

2) Optical Power Loss Mechanisms: The major loss processes in FSO propagation fall into four principal categories: geometric loss, atmospheric attenuation, atmospheric turbulence, and pointing errors.

Geometrical Loss. Geometric loss arises from beam divergence, which reduces the power density at the receiver. A commonly used approximation for uniformly illuminated circular apertures is [34]:

$$h_{\mathrm{geo}} = \left(\frac{D_{\mathrm{R}}}{D_T + R\,\theta} \right)^2, \tag{6}$$

where D_{R} denote the receiver aperture diameters (m). A typical FSO transceiver exhibits an optical divergence of 2–10 mrad for tracked systems and 0.05–1 mrad for untracked systems, corresponding to beam spreads between meters and tens of centimeters at a 1 km range [10].

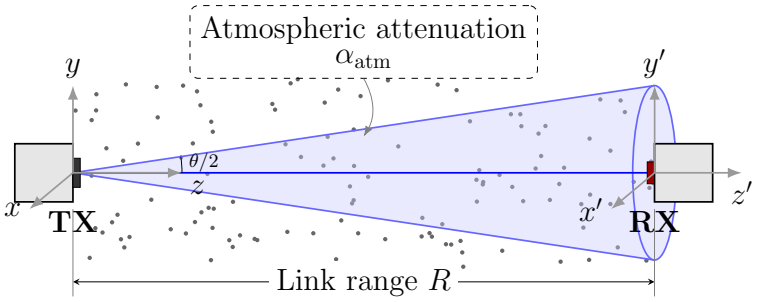
For Gaussian beams, geometric coupling is more accurately modeled using the collected fraction under *perfect alignment*. For a circular aperture of radius a (m) and a beam radius w_z (m) at R, the exact collected power fraction is given by [35]:

$$h_{\mathrm{geo}} = A_0 = \left[\mathrm{erf}\left(\frac{\sqrt{2}\, a}{w_z} \right) \right]^2, \tag{7}$$

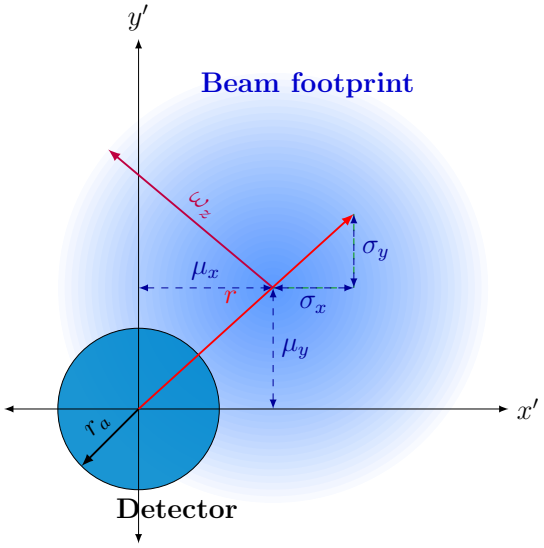
where $\mathrm{erf}(\cdot)$ is the error function. This expression represents the maximum geometric coupling efficiency when the beam is perfectly centered on the aperture, and already captures the impact of beam divergence through w_z. A mathematically convenient approximation of (7) can be obtained by approximating the error function expression with an equivalent exponential form [36]:

$$A_0 \simeq 1 - \exp\left(-\frac{2a^2}{w_z^2} \right), \tag{8}$$

which provides a convenient analytical approximation that closely matches the exact expression for typical aperture-to-beam radius ratios.



(a) Link geometry



(b) Beam footprint and detector misalignment

Fig. 4: FSO propagation model. (a) Optical wireless link geometry showing TX, RX, link range R, divergence angle θ, and atmospheric attenuation α_{atm}. (b) Beam footprint on the detector plane with misalignment characterized by mean offsets (μ_x, μ_y) and spreads (σ_x, σ_y).

Atmospheric Attenuation. Atmospheric attenuation is caused by absorption and scattering and is characterized by the Beer–Lambert Law [37]. The atmospheric transmittance is:

$$h_{\text{atm}} = \exp\big(-\alpha_{atm}\ R\big), \tag{9}$$

The corresponding loss in decibels is given by $-10\log_{10}(h_{\text{atm}}) \approx 4.343\,\alpha_{atm} R$.

Kim et al. [26] express α_{atm} in terms of V (km) and λ (nm) as:

$$\alpha_{atm} = \frac{3.91}{V}\left(\frac{\lambda}{550\text{ nm}}\right)^{-q(V)}, \tag{10}$$

with visibility-dependent exponent [38]:

$$q(V) = \begin{cases} 1.6, & V > 50 \text{ km}, \\ 1.3, & 6 < V < 50 \text{ km}, \\ 0.16V + 0.34, & 1 < V < 6 \text{ km}, \\ V - 0.5, & 0.5 < V < 1 \text{ km}, \\ 0, & V < 0.5 \text{ km}. \end{cases} \tag{11}$$

Table V summarizes visibility levels for representative weather conditions.

3) Time-Varying Channel Impairments: FSO links are also subject to dynamic channel impairments, particularly atmospheric turbulence and pointing errors.

Atmospheric Turbulence (Scintillation). Atmospheric turbulence originates from random temperature variations that create refractive index inhomogeneities in the propagation medium. As the optical beam propagates through these atmospheric eddies, it undergoes random amplitude and phase fluctuations, a phenomenon termed scintillation. The strength of these fluctuations is typically characterized by the C_n^2, which ranges between 10^{-17} m$^{-2/3}$ for weak turbulence and 10^{-13} m$^{-2/3}$ for strong turbulence [39]. The intensity fluctuation is quantified by the scintillation index:

$$\sigma_I^2 = \frac{\langle I^2\rangle - \langle I\rangle^2}{\langle I\rangle^2}, \tag{12}$$

where I denotes the instantaneous received optical intensity (W/m^2) at the receiver, and $\langle\cdot\rangle$ represents the ensemble average. Under the Rytov approximation for plane waves, the corresponding Rytov variance is defined as:

$$\sigma_R^2 = 1.23\, C_n^2\, n^{7/6} R^{11/6}, \tag{13}$$

where $n = 2\pi/\lambda$ is the optical wave number.

The statistical distributions commonly used to model these fluctuations include the *Lognormal distribution* for weak turbulence and the more versatile *Gamma–Gamma distribution*, which is valid across weak-to-strong regimes by separately modeling large-scale (α) and small-scale (β) fluctuations [39]:

$$f(h_{turb}) = \frac{2(\alpha\beta)^{(\alpha+\beta)/2}}{\Gamma(\alpha)\Gamma(\beta)}\, h_{turb}^{\frac{\alpha+\beta}{2}-1} K_{\alpha-\beta}\left(2\sqrt{\alpha\beta\, h_{turb}}\right), \tag{14}$$

for $h_{turb} > 0$ where $K_\nu(\cdot)$ is the modified Bessel function of the second kind.

As illustrated in Fig. 5, the structural behavior of the PDF shifts significantly as turbulence strength increases. Under weak turbulence, the distribution is narrow and centered near the normalized mean intensity of 1.0. However, as the channel enters the strong regime (indicated by decreasing α and β), the PDF becomes heavily skewed towards zero. This high probability density at low intensity levels represents the occurrence of deep signal fades, where the received power may fall below the detector sensitivity, leading to a surge in the BER.

While the PDF characterizes the temporal statistics of the received power at a point, the spatial manifestation of

TABLE IV: Description of symbols used in different optical wireless propagation environments.

Symbol	Description	Unit	Symbol	Description	Unit
Common notation across environments					
P_T	Transmitted optical power	W	P_R	Received optical power	W
R	Propagation distance / link range	m	λ	Optical wavelength	nm
Air (Atmospheric)					
A_{R}	Receiver aperture area	m^2	D_T, D_R	Transmitter/receiver aperture diameter	m
θ	Full beam divergence angle	rad	h_{geo}	Geometric spreading loss	–
h_{pe}	Pointing error loss	–	h_{atm}	Atmospheric transmittance	–
T	Optical transmission efficiency	–	K	Optical coupling factor	–
a	Receiver aperture radius	m	w_z	beam radius	m
C_n^2	Refractive index structure parameter	$\mathrm{m}^{-2/3}$	h_{turb}	Atmospheric turbulence loss factor	–
α, β	Gamma–Gamma turbulence parameters	–	σ_I^2	Scintillation index	–
n	Optical wave number	rad/m	α_{atm}	Atmospheric attenuation coefficient	m^{-1}
I, I_0	Instantaneous and mean optical intensity	–	σ_R^2	Rytov variance	–
μ_x, μ_y	Mean pointing offsets (x/y axes)	m	σ_x, σ_y	Std. deviation of pointing jitter (x/y axes)	m
V	Meteorological visibility	km	$q(V)$	Visibility-dependent exponent	–
σ_s	jitter standard deviation	-	$w_{z,eq}$	equivalent beam radius at the receiver plane	m
Φ	Pointing error angle	rad	δ	Vibration-induced angular deviation	rad
N_{TA}	Total noise power at transceiver A	W	Γ_{sA}	Residual self interference (SI) power at transceiver A	W
Indoor (VLC / IR OWC)					
d_o	Transmission distance	m	A_d	Photodiode active area	m^2
m	Lambertian emission order	–	$\phi_{1/2}$	Semi-angle at half power	rad
ϕ	Irradiance angle	rad	ω	Incidence angle	rad
ω_c	Receiver FoV	rad	n_i	Refractive index	–
$T_s(\omega)$	Optical filter gain	–	$g(\omega)$	Optical concentrator gain	–
ρ_k	Surface reflectivity of k-th reflector	–	ΔA	Emission area of reflecting element	m^2
k_B	Boltzmann's constant	J/K	T_K	Absolute temperature	K
η	Fixed capacitance of PD per unit area	m^2	B_n	Equivalent noise bandwidth	Hz
I_2, I_3	Noise bandwidth factors	–	G	Open-loop voltage gain	–
Γ	FET channel noise factor	–	g_m	FET Transconductance	A/V
e	Electronic charge	C	R_s	Responsivity of receiver	A/W
I_{bg}	Ambient light-induced background current	A	σ_{total}	Total noise variance	A^2
Underwater (UOWC)					
$a(\lambda)$	Absorption coefficient	m^{-1}	$b(\lambda)$	Scattering coefficient	m^{-1}
$c(\lambda)$	Extinction coefficient	m^{-1}	θ	Angle between the receiver plane normal and the Tx–Rx propagation path	rad
L_{pr}	Propagation loss factor	m^{-1}	h, x	Tx/Rx depth	m
ϕ	Incidence angle on receiver plane	degrees	$\theta_{\min}, \theta_{\max}$	Min/max reflection angles	degrees
η_T, η_R	Tx/Rx optical efficiencies	–	d	Propagation distance (Tx–Rx separation)	m
η_{Retro}	Retroreflector efficiency	–	$A_{\mathrm{Rec}}, A_{\mathrm{Retro}}$	Receiver/retroreflector areas	m^2
θ_0	Transmitter beam divergence angle	rad	$\theta_{0,\mathrm{Retro}}$	Retroreflector beam divergence angle	rad
Underground (IoUT / Soil / Mines)					
γ_n	Coupling space angle	rad	$n(t)$	AWGN	W
c	Speed of light	m^{-1}	f	Carrier frequency	Hz
I_N	Input current noise density	$\mathrm{A}/\sqrt{\mathrm{Hz}}$	G_B	Amplifier bandwidth	Hz
Space					
G_t, G_r	Transmitter and receiver aperture gain	–	η_{det}	Detector quantum efficiency	–
D_r	Receiver aperture diameter	m	γ_r	Obscuration ratio	–
L_{fs}	Free-space loss factor	–	L_a	Atmospheric transmittance factor	–
L_c	Cirrus-cloud transmittance	–	L_s	Scintillation loss factor	–
L_{pt}	Pointing-error loss factor	–	L_b, L_j	Blocking and jitter loss factors	–
$\theta_{1/e2}$	Beam divergence half-angle	rad	σ_p^2	Pointing error variance	rad^2
w_0	Beam waist radius	m	$P_{r,\mathrm{det}}$	Detected received power	W
$I(r, L)$	Irradiance at distance L	–	r	Radial displacement (misalignment)	m
$\bar{P}_r$	Mean received optical power	W	$\xi(t)$	Unit-mean random fading process (variance given by σ_I^2)	–
$P_{n,\mathrm{det}}$	Total detected noise power	W	$P_{r,\mathrm{ap}}$	Received power before detector	W

these fluctuations across the receiver aperture is illustrated in Fig. 6, providing a conceptual visualization of scintillation. The heatmap displays the normalized intensity where "hot spots" (red/yellow) indicate constructive interference (momentarily higher power) and "cold spots" (blue/green) indicate destructive interference or shadowing (local signal fades). This non-uniform spatial distribution is a fundamental cause of link unreliability. If a single, small receiver aperture happens to fall entirely within a localized cold spot, the system experiences an instantaneous outage.

Crucially, this visualization provides the intuitive basis for spatial diversity as a mitigation technique. By deploying multiple, separate receiver apertures, the system drastically lowers the probability that all sensors simultaneously fall into localized cold spots. This allows the receiver to effectively "average" the spatial fluctuations and harvest power from adjacent

TABLE V: Visibility range at different weather conditions.

Weather Condition	Visibility Range (km)
Thick Fog	0.20
Moderate Fog	0.50
Light Fog	0.77–1.00
Thin Fog/Heavy Rain (25 mm/h)	1.90–2.00
Haze/Medium Rain (12.50 mm/h)	2.80–4.00
Clear/Drizzle (0.25 mm/h)	18.00–20.00
Very Clear	23.00–50.00

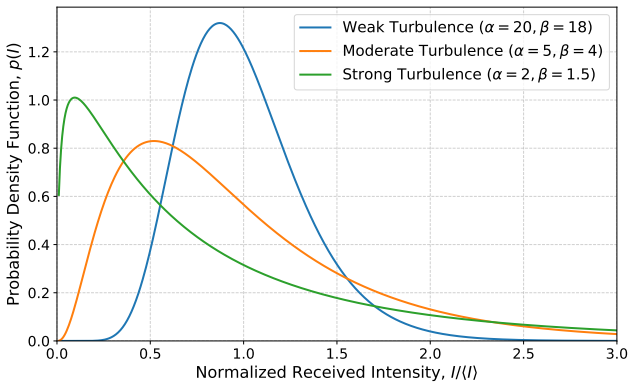


Fig. 5: Gamma-Gamma PDF under varying atmospheric turbulence regimes, showing the increased probability of deep fades as turbulence strength grows.

hot spots, significantly improving overall link availability and the system's fade margin.

TABLE VI: Comparison of pointing-error statistical models.

Model	Jitter Variance	Boresight/Drift Bias	Primary Use Case
Rayleigh	$\sigma_x = \sigma_y$	$\mu_x, \mu_y = 0$	Standard terrestrial links
Hoyt	$\sigma_x \neq \sigma_y$	$\mu_x, \mu_y = 0$	Asymmetric sway / UAV vibration
Rician	$\sigma_x = \sigma_y$	$\mu_x, \mu_y \neq 0$	Tracking errors / wind drift
Beckmann	$\sigma_x \neq \sigma_y$	$\mu_x, \mu_y \neq 0$	General mobile FSO scenarios

Pointing Errors (Jitter). Pointing errors arise from transmitter–receiver misalignment caused by mechanical vibration, airframe motion, or imperfect tracking. Let the lateral beam wander at the receiver be described by two orthogonal components (x, y), representing horizontal and vertical jitter. The radial displacement

$$r = \sqrt{x^2 + y^2}, \tag{15}$$

which determines the instantaneous misalignment, depends on the statistical properties (means and variances) of (x, y). Let, μ_x and μ_y denote the mean pointing offsets along the horizontal and vertical axes, respectively, while σ_x and σ_y represent the corresponding standard deviations of the pointing jitter, and σ_s denotes the common jitter standard deviation. In the following, we examine different statistical cases based on these parameters.

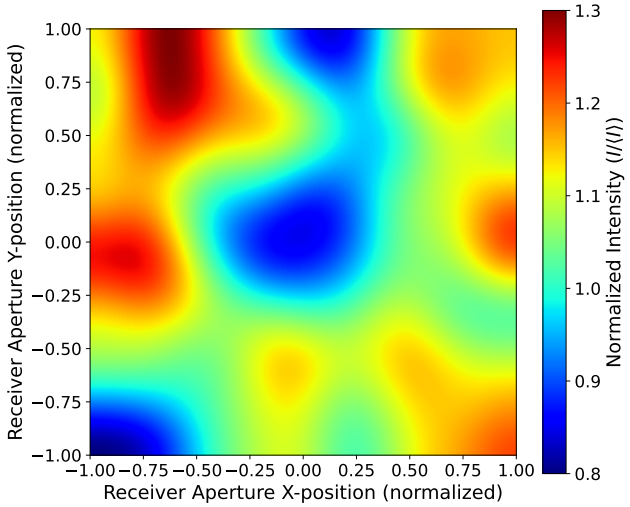


Fig. 6: Conceptual heatmap of instantaneous intensity distribution ($I/\langle I \rangle$) across the receiver aperture plane illustrating the spatial effects of scintillation.

Case 1: Equal variances, zero means ($\mu_x = \mu_y = 0,\ \sigma_x = \sigma_y = \sigma_s$)

When the jitter components are independent Gaussian variables,

$$x \sim \mathcal{N}(0, \sigma_s^2), \qquad y \sim \mathcal{N}(0, \sigma_s^2). \tag{16}$$

The radial displacement follows the classical Rayleigh distribution [35]:

$$f_r(r) = \frac{r}{\sigma_s^2} \exp\left(-\frac{r^2}{2\sigma_s^2}\right), \qquad r \geq 0. \tag{17}$$

This isotropic model is the widely used in FSO pointing-error analysis.

Case 2: Unequal variances, zero means ($\mu_x = \mu_y = 0,\ \sigma_x \neq \sigma_y$)

When horizontal and vertical jitter components have different variances, the vibration becomes anisotropic. In this case, the radial displacement follows a more general form often represented by the *Hoyt (Nakagami-q)* distribution, whose PDF is [40], [41], [42]:

$$f_r(r) = \frac{r}{q\,\sigma_y^2} \exp\left(-\frac{r^2(1+q^2)}{4q^2\sigma_y^2}\right) I_0\left(\frac{r^2(1-q^2)}{4q^2\sigma_y^2}\right), \tag{18}$$

where $q = \sigma_x/\sigma_y \in (0, 1]$ and $I_0(\cdot)$ is the modified Bessel function of the first kind. This model captures elliptical or axis-weighted scatter typical of UAV vibration patterns.

Case 3: Equal variances and nonzero means ($\sigma_x = \sigma_y,\ \mu_x \neq 0$ or $\mu_y \neq 0$)

When the horizontal and vertical jitter components have identical variances but exhibit a deterministic bias due to tracking errors or steady drift, the radial displacement follows the classical *Rician* distribution. The corresponding PDF is given by [43]

$$f_r(r) = \frac{r}{\sigma_s^2} \exp\left(-\frac{r^2+s^2}{2\sigma_s^2}\right) I_0\left(\frac{rs}{\sigma_s^2}\right), \qquad r \geq 0, \tag{19}$$

where $s = \sqrt{\mu_x^2 + \mu_y^2}$ represents the deterministic displacement magnitude. This model captures scenarios where the beam tends to drift toward a preferred direction due to imperfect tracking, wind-induced motion, or platform misalignment.

Case 4: Unequal variances and nonzero means ($\sigma_x \neq \sigma_y,\ \mu_x \neq 0$ or $\mu_y \neq 0$)

When jitter exhibits both anisotropy and directional bias (for example, persistent drift or wind-induced motion), the horizontal and vertical components (x, y) no longer correspond to the symmetric zero-mean models of Cases 1, 2 and 3. In this most general Gaussian setting, the pair (x, y) is often described by the *Beckmann model*, which extends the classical Rayleigh, Hoyt, and Rician families to the case of unequal variances and nonzero means. Its PDF is expressed in integral form as [44]:

$$f_r(r) = \frac{r}{2\pi\sigma_x\sigma_y}\int_0^{2\pi} \exp\left(-\frac{(r\cos\theta-\mu_x)^2}{2\sigma_x^2} - \frac{(r\sin\theta-\mu_y)^2}{2\sigma_y^2}\right)d\theta, \qquad r \geq 0. \tag{20}$$

While this general form lacks a simple closed-form solution, it is often approximated in recent literature to maintain analytical tractability for performance evaluation:

- Rayleigh-Approximated Beckmann: This model approximates the Beckmann distribution using a Rayleigh PDF modified by an adjustment parameter $\mathcal{G}$. This parameter is typically calculated using the third-order central moment of the beam deviation to provide a more tractable form for system analysis [44].
- Gamma-Approximated Beckmann: A more refined alternative that models the squared Beckmann distribution as a Gamma-distributed random variable. This approximation is noted for providing a near-perfect match to numerical results when evaluating metrics such as BER and outage probability [45].

Table VI summarizes the commonly used pointing-error statistical models and their corresponding assumptions and application scenarios.

Pointing-induced Power Loss. For Gaussian beams, the instantaneous pointing-error loss associated with a radial misalignment r can be approximated as [35]

$$h_{\text{pe}}(r) = \exp\left(-\frac{2r^2}{w_{z,eq}^2}\right), \tag{21}$$

where $w_{z,eq}$ denotes the equivalent beam radius at the receiver plane, which accounts for the effect of aperture truncation. Since r is a random variable due to pointing jitter, the loss factor h_{pe} is also random. Assuming that the horizontal and vertical pointing errors follow independent Gaussian distributions with variance σ_s^2, the radial displacement follows a Rayleigh distribution. Using the transformation $X = \exp(-2r^2/w_{z,eq}^2)$, the PDF of the normalized pointing-loss factor becomes

$$f_X(x) = \frac{w_{z,eq}^2}{4\sigma_s^2}\, x^{\frac{w_{z,eq}^2}{4\sigma_s^2}-1}, \qquad 0 < x \leq 1, \tag{22}$$

under the Rayleigh jitter model. Including geometric coupling, the total collected power fraction $H = A_0 X$ has PDF

$$f_H(h) = \frac{w_{z,eq}^2}{4\sigma_s^2 A_0}\left(\frac{h}{A_0}\right)^{\frac{w_{z,eq}^2}{4\sigma_s^2}-1}, \qquad 0 < h \leq A_0. \tag{23}$$

These expressions characterize the fading induced purely by pointing errors and are widely used to evaluate misalignment-driven outage probability in FSO systems. When the horizontal and vertical pointing jitters have unequal variances ($\sigma_x^2 \neq \sigma_y^2$), the radial displacement follows a Hoyt (Nakagami-q) distribution instead of the Rayleigh distribution. In this case, the pointing-loss statistics can be derived by averaging $h_{pe}(r)$ over the Hoyt-distributed radial displacement [42].

More generally, when the pointing jitter exhibits both anisotropy and directional bias, the radial displacement follows the Beckmann distribution. Since the corresponding pointing-loss PDF becomes mathematically intractable, several tractable approximations have been proposed in the literature, including the Rayleigh-approximated Beckmann model and the Gamma-approximated Beckmann model [44], [45]. A comparison between the statistical models is shown in Table VI.

4) Link Performance Modeling – SINR Formulations: With all attenuation mechanisms incorporated, the overall link performance is characterized using the SINR, which directly determines achievable data rates and link reliability.

Basic SINR formulation. SINR characterizes the quality of a communication system as well as it is the performance parameter for a transceiver. A fundamental expression for the SINR at transceiver A is given as [46]:

$$\text{SINR}_A = \frac{P_T\ \exp\left(-\alpha_{atm}R\right)\ \lambda^2\ A_{\text{R}}}{8\pi^2R^2(1-\cos\theta)(N_{TA}+\Gamma_{sA})}, \tag{24}$$

where N_{TA} (W) is the total noise power at transceiver A, typically computed from the noise-equivalent power (NEP) over the receiver bandwidth, and Γ_{sA} (W) is the residual SI power at transceiver A. Γ_{sA} is zero for half-duplex operation.

SINR with Pointing Errors and Vibrations. To account for platform instability, the model can include a pointing error angle Φ and a vibration angle δ [47]:

$$\text{SINR}_A = \left[\frac{P_T\ \exp\left(-\alpha_{atm}R\right)\ A_{\text{R}}\cos(\Phi+\delta)}{(\tan\theta)^2\ 4R^2(N_{TA}+\Gamma_{sA})}\right]^2. \tag{25}$$

SINR for Multi-Element Transceivers. For an advanced transceiver with N transmitters, the total SINR is the sum of contributions from each path [47], [48]:

$$\text{SINR}_A = \sum_{i=1}^{N}\sum_{j\in E_{T_i}}\left[\frac{P_{T,i}\exp\left(-\alpha_{atm}R\right)A_{\text{R}}\cos(\Phi_{ij}+\delta_j)}{(\tan\theta)^2 4R^2(N_{TA}+\Gamma_{sA})}\right]^2 \tag{26}$$

While physics-based analytical models provide fundamental insight into atmospheric propagation, their predictive validity depends strongly on turbulence strength, weather severity, wavelength regime, and receiver geometry. Under strong turbulence or highly scattering conditions, commonly used assumptions such as weak-fluctuation approximations or single-scattering attenuation may no longer hold. A comprehensive

discussion of these regime-dependent limitations is provided in Section XI.

C. Empirical Modeling in FSO Channel Characterization

Table VII provides a comprehensive summary of empirical FSO propagation models reported in the literature across different weather conditions, including their formulations, key parameters, and primary application scenarios. These models can be broadly grouped into three categories based on their functional objective.

Visibility-based attenuation models. These models estimate weather-induced losses, particularly under fog and haze, by relating optical attenuation to meteorological visibility and wavelength. Classical models by Kruse et al. [5] and Kim et al. [26], along with extensions by Al Naboulsi et al. [6] and Ijaz et al. [49], refine the visibility exponent ($q(V)$) to account for particle size distributions and fog subtypes (e.g., advection versus convection). Due to their simplicity and interpretability, they remain the standard for initial link budget planning.

Statistical fading models. These models characterize turbulence-induced irradiance fluctuations using received power statistics. Works such as Epple [51] utilize metrics like the scintillation index to fit log-normal or Gamma-based distributions. These models are primarily used to generate simulation-ready channel traces for evaluating modulation, coding, and diversity techniques under short-term fading.

Regression-based empirical models. More recent approaches employ low-dimensional parametric regression to unify measurements across multiple locations. For example, Esmail *et al.* [50] derive nonlinear regression laws that significantly reduce prediction error compared to classical visibility-based models, while enabling probabilistic analysis through fitted PDFs.

A comparative analysis of these empirical approaches reveals that while classical visibility-based models (Kruse/Kim) offer global applicability, they frequently overestimate link performance in low-visibility regimes ($V < 1$ km). Conversely, subtype-specific models (Al Naboulsi) provide higher accuracy but require precise meteorological classification of fog types, which is often unavailable in real-time deployments. The recent trend in research indicates a shift toward regression-based models that prioritize local calibration over global generalization, acknowledging that the "best" empirical fit is highly dependent on regional aerosol characteristics.

While empirical models provide an essential bridge to real-world deployment through interpretability and ease of use, their reliance on fixed functional forms limits their ability to capture complex, multi-parameter environmental interactions. This limitation motivates the transition to high-dimensional, data-driven ML approaches, discussed in the following section.

D. ML–Based Propagation Models

As summarized in Table VIII, ML-based propagation modeling efforts can be grouped into four broad categories based on the modeling objective.

Link-metric regression from environmental features. These works learn direct mappings from meteorological or environmental measurements to link metrics such as received power or turbulence strength. For example, long-term maritime monitoring can be formulated as regression from weather inputs to RSSI, while other studies predict turbulence-related quantities (e.g., $\log_{10} C_n^2$) from routine sensor measurements [30], [52]. Such models are attractive for network planning and monitoring because they can leverage readily available sensor data and provide fast predictions once trained.

Availability/outage prediction and classification. Here, ML is used to predict whether the link is functional under a specified availability target, often using a physics-based outage definition to generate labels (or using measured availability labels). This framing naturally supports binary classification (functional/non-functional) and is useful for proactive switching, scheduling, or hybrid RF/FSO control [52].

Sensorless parameter monitoring from received-signal features. Instead of relying solely on external sensors, some studies extract features from the received signal (e.g., asynchronous histograms) and use ML to estimate latent channel parameters such as OSNR, turbulence strength, and pointing-error severity [56]. These approaches aim to enable real-time monitoring without dedicated instrumentation, but their generalization depends on how representative the training data are of the deployment conditions.

Learning-assisted detection and control under propagation impairments. ML has also been applied at the receiver or control layer, including learning-based detection/decoding that reduces reliance on explicit channel state information (CSI), and learning-based pointing/beam control in dynamic scenarios [53]–[55]. These works do not replace propagation physics, but instead learn decision rules (detection boundaries or control actions) that are robust to propagation variability.

Across these categories, ML models offer *flexibility* in incorporating many correlated inputs and *speed* in inference after training, which is particularly useful for monitoring and adaptation. However, ML-based propagation modeling is inherently data-dependent: prediction quality hinges on the coverage of weather regimes, deployment geometry, and hardware configurations in the training set. As a result, careful validation, domain-shift awareness, and (when possible) physics-guided features or hybrid modeling can be important to ensure that ML predictions remain reliable outside the specific conditions used for training [31].

V. Indoor Propagation Models

Beyond propagation in free-space, indoor environments present their own kind of issues, mainly due to multipath effects and unavailability of LoS.

A. Propagation Challenges

In indoor environments, the following challenges are prominent for OWC:

- Optical wireless channels do not experience small-scale fading due to multipath interference, and Doppler shift

TABLE VII: Summary of empirical FSO propagation models across weather conditions.

Model Family	Environment	Formula / Derived Relationship	Key Parameters	Primary Focus
Kruse visibility model [5]	Fog/Haze (Atmospheric)	$\alpha_{\text{atm}} = \frac{13}{V}\left(\frac{\lambda}{0.55}\right)^{-q(V)}$, $q(V) = \begin{cases} 1.6, & V > 50 \\ 1.3, & 6 < V < 50 \\ 0.585\,V^{1/3}, & V < 6 \end{cases}$	V, λ	Baseline visibility-based fog attenuation
Kim visibility model [26]	Fog/Haze (Atmospheric)	$\alpha_{\text{atm}} = \frac{13}{V}\left(\frac{\lambda}{0.55}\right)^{-q(V)}$, $q(V) = \begin{cases} 1.6, & V > 50 \\ 1.3, & 6 < V < 50 \\ 0.16V + 0.34, & 1 < V < 6 \\ V - 0.5, & 0.5 < V < 1 \\ 0, & V < 0.5 \end{cases}$	V, λ	Low-visibility correction to Kruse model
Al Naboulsi fog model (Convection) [6]	Fog (Convection)	$\alpha_{\text{atm}} = 4.343 \cdot \frac{0.11478\lambda + 3.8367}{V}$	V, λ	Convection fog attenuation modeling
Al Naboulsi fog model (Advection) [6]	Fog (Advection)	$\alpha_{\text{atm}} = 4.343 \cdot \frac{0.18126\lambda^2 + 0.13709\lambda + 3.7502}{V}$	V, λ	Advection fog attenuation modeling
Ijaz controlled fog model [49]	Fog (Controlled chamber)	$\alpha_{\text{atm}} = \frac{17}{V}\left(\frac{\lambda}{0.55}\right)^{-q(\lambda)}$, $q(\lambda) = 0.1428\lambda - 0.0947$	λ (Fog chamber)	Experimental fog attenuation validation
Esmail regression model [50]	Fog (Multi-site)	$\alpha_{\text{atm}} = \mathcal{C} \cdot V^{-(a\lambda+b)}$	V, λ (Multi-city data)	Unified regression-based fog attenuation
Epple statistical model [51]	Atmospheric turbulence	$\sigma_P^2 = \frac{\mathbb{E}[P_{\text{R}}^2]}{\mathbb{E}[P_{\text{R}}]^2} - 1$ (Log-normal / Gamma fits)	Received power stats (PDF fitting)	Statistical scintillation and fading trace generation

has negligible impact on optical wireless [57]. However, it suffers from multipath dispersion, large-scale fading due to high path loss and shadowing that can cause inter-symbol interference (ISI) [11], [58].

- Optical signal reception can get affected by incandescent-light and fluorescent-light noises and shot noise caused by background ambient light, which requires high-pass optical filtering to achieve high SNR [57], [59], [60].
- Optical signals can withstand attenuation from absorption and scattering, however, get affected by surface reflectance and blockage [61]–[63].
- Optical wireless systems can suffer from co-channel interference caused by the reuse of the same optical spectrum [64].
- Antenna orientation and user mobility can impact the LoS link geometry and cause substantial SNR drop [65].

B. Link Configurations

Based on different specifications such as directionality of optical transmitter, existence of LoS path between transmitter and receiver, transmitter divergence angle and receiver Field-of-View (FoV), optical wireless channel link configurations for indoor environment can be categorized as [11]:

1) *Directed LoS:* Suitable for all indoor OWC technologies, specially in WIRC operating between 850 nm and 950 nm wavelengths, with wireless LoS link ranging up to 50 m. Because of its short-range highly directional communication characteristics, it can maximize power efficiency and thus attain high data rate with low power requirement. However, such strict link alignment between communication devices results to small coverage area and limited mobility, and can easily get affected by shadowing and blockage.
2) *Non-Directed LoS:* Due to its tolerance to misalignments, non-directed LoS link allows point-to-multipoint communication using wide optical aperture of receivers, making it suitable for indoor environments where mobility is involved. It can withstand the impacts of shadowing and blockage, however, shows sensitivity to reflections and multipath dispersion leading to high path loss and constrained bandwidth, respectively. Due to the capability to establish multi-access links using same source, it enables the use of VLC technology in indoor environments.
3) *Diffused:* Uses WIRC for indoor setup. No LoS required. Transmitter with wide beamwidth transmits towards the ceiling. The signal gets diffused and reflected off ceilings, walls, and objects. Receivers with wide aperture receive the signal while facing the ceiling. Handles user mobility and withstands link loss due to blockage. However, sustains high path loss due to no LoS, along with multipath dispersion which causes ISI. Less power efficient because of fraction of total power is used to tackle ISI. For outdoor setup, wireless ultraviolet communications (WUVC) is used with more distance between Tx-Rx.
4) *Tracked:* In a single link configuration, tracked links can deliver high data rates with decreased multipath-induced

TABLE VIII: Summary of ML-based propagation modeling approaches in FSO/OWC systems.

Model	Environment	Learned Relationship / Task	Key Inputs	Focus
[30]	Maritime atmosphere (2.96 km link, Piraeus port, Greece)	Supervised regression: meteorological feature vector → received power (RSSI); models compared (KNN/DT/random forest/GBR/ANN)	T, RH, WS, SF, DP, P, ASTD	Received power prediction for monitoring/planning
[52]	Terrestrial near-maritime atmosphere (NPS rooftop)	Regression: meteorological features → $\log_{10} C_n^2$; classification: outage/availability label; plus polynomial surrogate for P_{out}	T_{air}, U, H_2O, SF, P_{air}, T_g, C_n^2	Availability/outage prediction and turbulence regression
[53]	Satellite-to-ground FSO (simulated)	Learning-assisted control: Conv1D predicts control gain to reduce pointing error; evaluated via outage under log-normal turbulence	Simulated state/measurement sequences	Pointing-error mitigation/control
[54]	Atmospheric FSO (simulated impairments)	SVM-based symbol detection for OOK without explicit CSI; BER evaluation under modeled impairments	Received samples (labeled) + scenario parameters	Low-complexity detection without CSI
[55]	Atmospheric FSO (simulated: weak–saturated turbulence)	DNN-based OOK detection: received vector → bit estimates; compared to CSI/blind ML detection	Received sample vectors (labeled)	Learning-based detection under turbulence
[56]	Atmospheric FSO (simulated, 1 km OOK, 1550 nm)	Regression from asynchronous histograms → OSNR, C_n^2, pointing severity ζ; SVR vs CNN	AAH/ADTS histogram features	Sensorless monitoring of channel/impairments

ISI, increased power efficiency under user mobility. A tracked system can be integrated into a mobile terminal, a base station, or a transceiver installed on the ceiling. Three primary components make up the tracking mechanism: a pointing unit to aim the optical beam so that it is on or nearly parallel to the necessary optical path, a tracking unit to track the new mobile station, and an acquisition unit to detect the presence of a new mobile station within a given area. Tracked link configuration is more practical in indoor WIRCs.

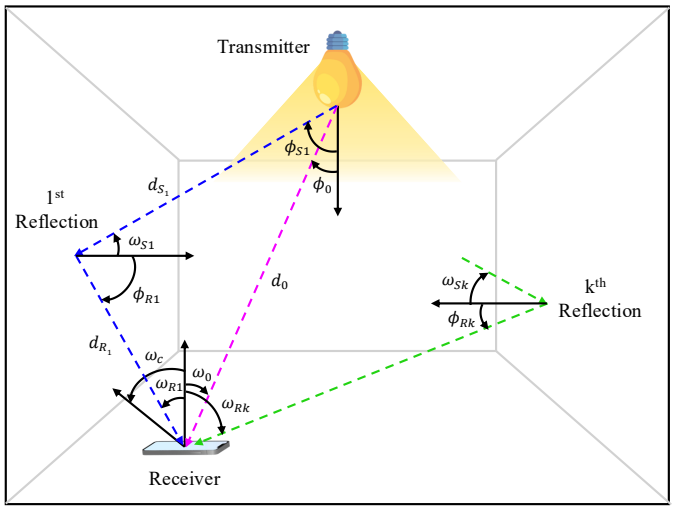


Fig. 7: Geometry used to describe analytical modeling of indoor propagation channel.

C. Analytical Modeling

The order of Lambertain emission of a light emitting diode (LED), denoted by m, can be expressed as [66]:

$$m = -\frac{\ln 2}{\ln(\cos \phi_{1/2})}, \tag{27}$$

where $\phi_{1/2}$ is the LED semi-angle at half optical power (rad).

The generalized Lambertian radiant intensity as a function of irradiance angle ϕ (rad) can be expressed as [67]:

$$R(\phi) = \begin{cases} \frac{m+1}{2\pi} \cos^m(\phi), & \phi \in [-\pi/2, \pi/2] \\ 0, & \phi \geq \pi/2. \end{cases} \tag{28}$$

Given the internal refractive index n_i, the optical gain of an ideal concentrator as a function of incidence angle ω (rad) can be expressed as [68]:

$$g(\omega) = \begin{cases} \frac{n_i^2}{\sin^2 \omega_c}, & 0 \leq \omega \leq \omega_c \\ 0, & \omega \geq \omega_c, \end{cases} \tag{29}$$

where ω_c is the RX FoV.

For *Directed LoS* and *Tracked* links in an indoor setup between a single Lambertian source TX (x, y, z) and a receiver RX (x', y', z') positioned at distance R (m), as depicted in Fig. 7, where RX is configured with an optical band-pass filter of gain $T_s(\omega)$, with irradiance angle ϕ (rad), incidence angle ω (rad), concentrator gain $g(\omega)$, the channel direct current (DC) gain of the receiver side can be expressed as [69]:

$$H_{\text{LoS}}(0) = \begin{cases} \frac{A_{\text{R}}}{R^2} R(\phi) \cos(\omega) T_s(\omega) g(\omega), & 0 \leq \omega \leq \omega_c \\ 0, & \omega > \omega_c. \end{cases} \tag{30}$$

The corresponding received power $P_{\text{R,LoS}}$ (W) can be defined by [70]:

$$P_{\text{R,LoS}} = P_T H_{\text{LoS}}(0). \tag{31}$$

For *Non-Directed LoS* and *Diffused* links between the same source TX and RX, CIR can be expressed by [71]:

$$h_{\text{NLoS}}(t; \text{TX}, \text{RX}) = \sum_{k=0}^{\infty} h_{\text{NLoS}}^{(k)}(t; \text{TX}, \text{RX}), \tag{32}$$

where $h_{\text{NLoS}}^{(k)}(t; \text{TX}, \text{RX})$ is the component channel impulse ratio (CIR) where signals from TX undergo k reflections

before reaching RX. Considering finite amount of reflections K, Equation (32) can be re-written as:

$$h_{\mathrm{NLoS}}(t;\mathrm{TX},\mathrm{RX}) = \sum_{k=0}^{K} h_{\mathrm{NLoS}}^{(k)}(t;\mathrm{TX},\mathrm{RX}), \tag{33}$$

which can be analyzed using a recursive algorithm or iterative methods [12]:

$$h_{\mathrm{NLoS}}^{K}(t;\mathrm{TX},\mathrm{RX}) = \sum_{k=1}^{K} R_o(\phi_k)\,\rho_k\,\frac{\cos(\omega)}{d_{\mathrm{TX}-k}^2}\,\mathrm{rect}\left(\frac{2\omega}{\pi}\right) \times\; h_{\mathrm{NLoS}}^{k-1}\left(t-\frac{d_{\mathrm{TX}-k}}{c},\,\mathrm{TX},\,\mathrm{RX}\right)\,\Delta A, \tag{34}$$

where ΔA is the emission area of a reflecting element (m^2), $R_o(\cdot)$ is the radiant intensity given irradiance angle ϕ_k, ρ_k is the surface reflectivity of k-th reflecting element, $d_{\mathrm{TX}-k}$ is the distance from TX to reflector k, $h_{\mathrm{NLoS}}^{k-1}(t,\mathrm{TX},\mathrm{RX})$ denotes the impulse response of order $k-1$ between RX and reflector k, and rect$(\cdot)$ can be expressed by [72]:

$$\mathrm{rect}(x) = \begin{cases} 1, & |x| \le 1 \\ 0, & |x| > 1, \end{cases} \tag{35}$$

where $|\cdot|$ represents cardinality of a set.

Using the single-reflection propagation the channel DC gain for the first diffuse reflection can be approximated as [73]:

$$dH_{\mathrm{NLoS}}^{1}(0) = \begin{cases} \frac{A_{\mathrm{R}}}{d_{\mathrm{TX}-k}^2 d_{\mathrm{RX}-k}^2} R_o(\phi_{\mathrm{TX}-k})\rho_k \Delta A \cos(\omega_{\mathrm{TX}-k}) \\ \times \cos(\phi_{k-\mathrm{RX}})\cos(\omega_{k-\mathrm{RX}}) \\ \times T_s(\omega_{k-\mathrm{RX}})g(\omega_{k-\mathrm{RX}}), & 0 \le \omega \le \omega_c \\ 0, & \omega > \omega_c, \end{cases} \tag{36}$$

where $d_{\mathrm{TX}-k}$ is the distance between TX and reflecting element k (m), $d_{k-\mathrm{RX}}$ is the distance between the reflecting element k and RX (m), $\omega_{\mathrm{TX}-k}$ denotes the incidence angle of the reflecting element k (rad), $\phi_{\mathrm{RX}-k}$ denotes radiation angle of the receiver (rad), and $\omega_{k-\mathrm{RX}}$ denotes the incidence angle of the receiver RX (rad).

Total received optical power in association with the optical transmission power P_T (W), channel DC gains of LoS and reflected NLoS links can be defined by:

$$P_{\mathrm{R,total}} = P_T\Big[H_{\mathrm{LoS}}(0) + \int dH_{\mathrm{NLoS}}^{1}(0)\Big]. \tag{37}$$

Along with the total received power $P_{\mathrm{R,total}}$ (W), RX experiences additive noise N (W), consisting of shot noise and thermal noise, which can be represented by Additive White Gaussian Noise (AWGN) with zero mean and variance $\sigma_{\mathrm{total}}^2$ as [74]:

$$\sigma_{\mathrm{total}}^2 = \sigma_{\mathrm{thermal}}^2 + \sigma_{\mathrm{shot}}^2. \tag{38}$$

The variances of different noise components can be expressed as [75]:

$$\sigma_{\mathrm{thermal}}^2 = 8\pi k_B T_K \eta A_d B_n^2\Big(\frac{I_2}{G} + \frac{2\pi\Gamma}{g_m}\eta A_{\mathrm{R}} I_3 B_n\Big) \tag{39}$$

$$\sigma_{\mathrm{shot}}^2 = 2eB_n(R_s P_{\mathrm{R,total}} + I_{\mathrm{bg}} I_2), \tag{40}$$

where k_B is the Boltzmann's constant that is $1.380649 \times 10^{-23}\mathrm{J/K}$, T_K is the absolute temperature (K), η is the fixed capacitance of RX per unit area ($\mathrm{F/m}^2$), B_n is the equivalent noise bandwidth (Hz), I_2 and I_3 are the noise bandwidth factors where $I_2 = 0.562$ and $I_3 = 0.0868$, G denotes the open-loop voltage gain, Γ is the Field-effect Transistor (FET) channel noise factor, g_m is the FET transconductance (A/V), e is the electronic charge (C), R_s is the responsivity of the receiver (A/W), and I_{bg} is the ambient light-induced background current (A).

The SNR at receiver end can be computed by [76]:

$$\mathrm{SNR} = \frac{(R_s P_{\mathrm{R,total}})^2}{\sigma_{\mathrm{total}}^2}. \tag{41}$$

In addition to noise N, receiver may also experience ISI. Depending of the design of LED source, ISI can be generated by an optical path difference either from the reflection or from both LoS and NLoS propagations [77]. Considering signals arriving after the symbol period T, the ISI-induced received power can be expressed by [78]:

$$P_{\mathrm{R,ISI}} = \int_{\mathrm{T}}^{\infty} \big[x(t) \otimes h(t)\big], \tag{42}$$

where $x(t)$ and $h(t)$ denote the transmitted optical signal and the corresponding CIR, respectively. $\otimes$ denotes convolution operation.

Considering $P_{\mathrm{R,ISI}}$, the noises in equations (38) and (40) can be modified to [79]:

$$(\sigma_{\mathrm{shot}}^*)^2 = 2eR_s B_n(P_{\mathrm{R,total}} + P_{\mathrm{R,ISI}}) + 2eI_{\mathrm{bg}} I_2 B_n \tag{43}$$

$$\sigma_{\mathrm{total}}^2 = \sigma_{\mathrm{thermal}}^2 + (\sigma_{\mathrm{shot}}^*)^2 + (R_s P_{\mathrm{R,ISI}})^2. \tag{44}$$

D. Empirical Modeling

Simulation-based deterministic models. These approaches allow explicit modeling of room geometry, transmitter and receiver characteristics, and multi-bounce reflections to generate CIRs. Fundamental models like recursive algorithm [80], along with VLC-specific extensions [81], enable practical estimation of power-delay profiles for indoor VLC channels. Ray-tracing techniques [82] further improve the modeling precision and are frequently employed to implement standardized indoor use cases, including the IEEE reference channel models [83].

Regression-based empirical models. Regression-based models apply compact, site-calibrated parametric laws to measurements or high-precision simulation datasets for rapid evaluation with preserved interpretability. The typical workflow follows initial channel data generation using advanced ray-tracing, then compression of resulting propagation behavior into closed-form expression for path loss or shadowing via curve-fitting [84], [85]. Parameters of such fitted model exhibit strong dependency on site-specific factors and can vary with surface reflectance, room size and material properties.

Statistical variability models. Performance of indoor VLC often gets affected by dynamic channel impairments like receiver pose dynamics, receiver orientation and mobility, FoV constraints, and intermittent human blockage and shadowing. Accordingly, these modeling techniques treat the channel gain

or LoS availability of such environment as random variables and derive gain distributions and LoS availability to support reliability analysis [86]–[88].

A comparative analysis of these empirical approaches indicates that simulation-based deterministic models yield the highest-precision CIRs and are optimal for studying multipath dispersion and site-specific performance bounds. However, they necessitate intricate environmental specifications and entail substantial computational expenses, thereby constraining scalability beyond standardized scenarios [83]. Statistical variability models are crucial for reliability-driven design as they directly address the key factors behind VLC intermittency. However, scenario dependency and oversimplified statistical assumptions can limit their applicability, perhaps failing to properly reflect the correlated, time-varying interactions among orientation and blockage dynamics across diverse indoor setups. Regression-based approaches offer the most practical overall trade-off, enabling fast link-budget evaluation with strong accuracy by prioritizing site-specific calibration.

E. ML-based Models

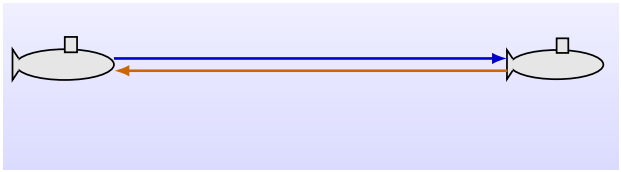
In [89], a low-complexity, Neural Network (NN) based adaptive channel estimator is proposed to tackle frequency-selective fading in dispersive OWC channels for indoor settings. Using ray tracing simulator, an indoor setting with a room of dimensions $5m \times 5m \times 5m$ is modeled with single Transmitter/Receiver, coefficient of reflection (ρ) 0.7, 45° FoV at both ends, and the SNR is varied in range 15dB to 30dB to capture the environment geometry. An uncoded DCO-OFDM system is employed with subcarriers N_f = 14, cyclic prefix with length L_{CP} = 7, pilot spacing L_s = 5, DC bias 7dB with 30KHz frequency spacing. For experiment, a two-path indoor channel with one specular reflection along with LoS link is considered to produce weak frequency responses. CIR is used as the key feature for the proposed framework. Preprocessing starts by forming the LS estimate $\hat{H}^{\text{LS}}$ from the received pilot signal Y_p.

The paper uses InterpolateNet [90], a lightweight CNN with 9,442 tunable parameters, that maps the channel estimate $H_{DNN} = \mathcal{F}_P(H_{LS})$ for certain set parameters P. Three different InterpolateNet networks are initiated to train on channel samples for low delay spread (LDS), medium delay spread (MDS), and high delay spread (HDS). Initial CIR estimation h_{est} is completed using P_{HDS}, which is then compared against h_{HDS} and h_{LDS} to the best model to use for more precise estimation.

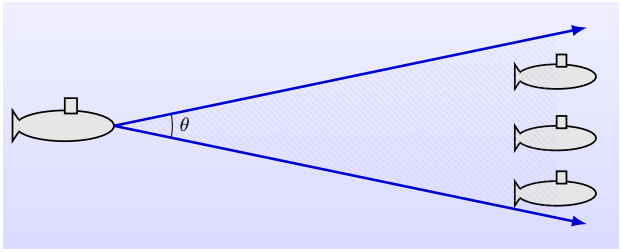
The three InterpolateNet networks collectively use an offline training dataset of 100,000 channel samples with 95%-5% split. For evaluation, Normalized mean square error (NMSE) and BER are used as key performance metrics. The proposed method consistently outperforms LS and single HDS-trained estimates, achieving 1.5% and 1.04% reductions in NMSE, and 20% and 11.71% reductions in BER at 20dB SNR, respectively.

VI. Underwater Propagation Models

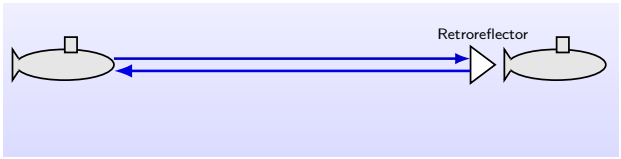
Underwater wireless communications (UWCs) involve the transmission of data in unguided aquatic environments through various wireless carriers, such as RF, acoustic, and optical waves. Traditionally, UWC utilizes acoustic waves due to their low loss in water, allowing for long-range communication [92]–[94]. However, acoustic communication has limitations, such as a narrow bandwidth resulting in low bit rates and a slow wave speed of 1500 m/s, which causes latency. To achieve higher bit rates, optical frequencies are considered a promising alternative. Underwater OWC can reach up to 425 m, which is sufficient for unmanned underwater vehicles (UUVs) and divers in focused scenarios like search and rescue operations where real-time, high-quality images are essential [95], [96]. Thus, high-rate data transmission between underwater nodes and the buoy is necessary for these applications. Compared to RF and acoustic methods, UOWC offers significantly higher transmission bandwidth and data rates [91].



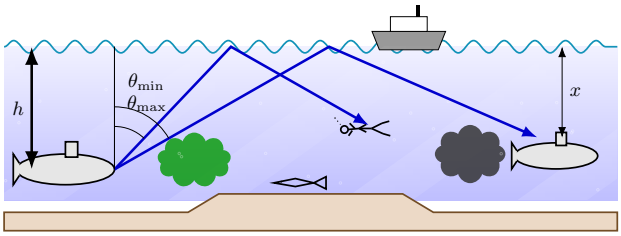
(a) Point-to-point LoS configuration



(b) Diffused LoS configuration



(c) Retroreflector-based LoS



(d) NLoS configuration

Fig. 8: UOWC link configurations [17], [91].

However, UOWC faces several significant challenges:

- Absorption and scattering of light by water molecules and

particles drastically reduce the intensity of the signal, limiting the effective communication range [95]. Scattering, in particular, creates dispersion and introduces ISI, which is more pronounced at high bit rates [96].
- Turbulence within the water and at the water surface, caused by variations in temperature, salinity, and wind, can lead to signal fading and fluctuations (scintillation), further complicating reliable data transmission [95], [97].
- The quality of the water plays a critical role; particulates and dissolved substances adversely affect signal propagation by introducing additional scattering and absorption [95].

A. Optimal Wavelength for Underwater OWC

The performance of a UOWC link is highly dependent on the choice of operating wavelength, which is dictated by the optical properties of the specific water environment. Water exhibits a low-attenuation "transmission window" in the blue-green portion of the visible spectrum [96]. The ideal wavelength shifts based on the concentration of organic matter, such as chlorophyll, and dissolved substances like humic and fulvic acids, collectively known as colored dissolved organic matter (CDOM) [95]. In general, the followings hold:

- In pure sea or clear ocean water, where concentrations of these substances are low, the minimum attenuation window is in the blue-green region (450–500 nm).
- In coastal or turbid harbor waters, higher concentrations of organic matter cause increased absorption in the blue region, shifting the optimal transmission window to the yellow-green part of the spectrum (520–570 nm) [95].

Table IX summarizes the optimal transmission wavelengths for different water types [18].

TABLE IX: Optimal wavelengths for various water types [18].

Water Type	**Chlorophyll Concentration**	**Humic & Fulvic Acid Level**	**Operating Wavelength**
Clear seawater	Low	Low	450–500 nm (blue-green)
Coastal waters	High	High	520–570 nm (yellow-green)
Turbid harbor water	Very high	Very high	520–570 nm (yellow-green)*

*Limited by absorption and scattering.

B. Analytical Modeling

Modeling the UOWC channel is essential for predicting link performance and designing system parameters. The underwater propagation of light is affected by attenuation from absorption and scattering, geometric losses, and in multi-node scenarios, interference. Analytical models provide a foundational framework for estimating link budgets. The key parameters used in these models are defined in Table IV.

Optical Attenuation Model. The attenuation of an optical signal in water is primarily described by the extinction coefficient $c(\lambda)$, which is the sum of the absorption coefficient $a(\lambda)$ and the scattering coefficient $b(\lambda)$ [95]:

$$c(\lambda) = a(\lambda) + b(\lambda). \tag{45}$$

Based on the Beer-Lambert law, the propagation loss factor L_{pr} over a distance R is an exponential function of the extinction coefficient [17]:

$$L_{pr}(\lambda, R) = \exp[-c(\lambda)R]. \tag{46}$$

Link Configuration Models. Different link configurations can be employed to establish a UOWC connection as shown in Fig. 8, each with a distinct mathematical model for the received power (P_{R}) [17].

1) *LoS Link*: This is the most straightforward configuration, where the transmitter points directly at the receiver. The received power is given by [17]:

$$P_{\mathrm{R,LoS}} = P_T\ \eta_T\ \eta_R\ L_{pr}\left(\lambda, \frac{R}{\cos\Phi}\right) \times \frac{A_R \cos\Phi}{2\pi R^2(1-\cos(\theta/2))}, \tag{47}$$

where η_T, η_R are optical efficiencies of the transmitter and receiver, respectively.

2) *Modulating Retroreflector Link*: This configuration is useful in asymmetric scenarios. A powerful interrogator illuminates a modulating retroreflector, which reflects the signal back with encoded data. The received power for this two-way path is [17]:

$$P_{\mathrm{R,Retro}} = P_T\,\eta_T\eta_R\eta_{\mathrm{Retro}}\,L_{\mathrm{pr}}\big(\lambda, \tfrac{2R}{\cos\Phi}\big) \times \left[\frac{A_{\mathrm{Retro}}\cos\Phi}{2\pi R^2(1-\cos(\theta/2))}\right]\left[\frac{A_{\mathrm{R}}\cos\Phi}{\pi(R\tan\theta_{\mathrm{retro}})^2}\right], \tag{48}$$

where η_{Retro} is the optical efficiency of the retroreflector, A_{Retro} is the retroreflector aperture area (m^2), and θ_{Retro} is the retroreflector beam divergence angle (rad). The term $L_{pr}(\lambda, \cdot)$ accounts for propagation loss as a function of wavelength and path length, where the argument $\frac{2R}{\cos\Phi}$ represents the round-trip propagation distance. The two geometric terms correspond to the transmitter-to-retroreflector and retroreflector-to-receiver coupling losses, respectively.

3) *Reflective Link (NLoS)*: When a direct LoS path is obstructed, the water-air interface can be used as a reflector to establish a NLoS link. In this case, the laser emits an optical beam forming a cone of light, defined by the inner and outer angles $\theta_{\min}$ and $\theta_{\max}$, respectively, in the upward direction, as illustrated in Fig. 8(d). The received power is approximated by [17]:

$$P_{\mathrm{R,ref}}(\Phi) \approx A_{\mathrm{Rec}} f_{\mathrm{R,ref}}(\Phi), \tag{49}$$

where the auxiliary function $f_{\mathrm{R,ref}}(\Phi)$ depends on whether the angle of incidence is below or above the critical angle for total internal reflection [17] and can be expressed as:

$$f_{R,\text{ref}}(\Phi) = \begin{cases} \dfrac{P_T \eta_T \eta_R L_{\text{pr}}(\lambda, h+x)\cos\Phi}{2\pi(h+x)^3(\cos\theta_{\min}-\cos\theta_{\max})} \\ \times \left[\frac{1}{2}\left(\frac{\tan(\theta_t-\Phi)}{\tan(\theta_t+\Phi)}\right)^2 + \left(\frac{\sin(\Phi-\theta_t)}{\sin(\Phi+\theta_t)}\right)^2\right], \theta_{\min} \le \Phi \le \theta_c, \\ \dfrac{P_T \eta_T \eta_R L_{\text{pr}}(\lambda, h+x)\cos\Phi}{2\pi(h+x)^3(\cos\theta_{\min}-\cos\theta_{\max})}, \theta_c < \Phi \le \theta_{\max}. \end{cases} \tag{50}$$

Here, h and x are transmitter and receiver depth from sea surface, respectively, θ_t denote the angle of transmission, obtained using Snell's law. This model accounts for the reflection from the ocean surface and depends on incident and transmission angles. Each of the three analytical models presented is suitable for specific underwater communication scenarios. The LoS model offers the highest power efficiency but requires alignment. The retroreflector model supports asymmetric energy profiles but suffers from doubled attenuation. The reflective model enables communication without direct LoS, trading off power efficiency for robustness.

C. Channel Performance Metrics: SNR and SINR

1) SNR: For point-to-point UOWC links where interference is negligible, the SNR is a key performance metric [96]. The electrical SNR can be expressed as [98]:

$$\text{SNR} = \left[\frac{P_T \ \exp\big(-c(\lambda)R\big) D_R^2 \cos\Phi}{(\tan^2\theta) 4R^2 N_T}\right]^2, \tag{51}$$

where, N_T is calculated as [98]:

$$N_T = \sqrt{P_{\text{bg_sn}}^2 + P_{\text{sig_sn}}^2 + P_{\text{dark_sn}}^2 + P_{\text{amp_n}}^2}. \tag{52}$$

Here, $P_{\text{bg_sn}}$, $P_{\text{sig_sn}}$, $P_{\text{dark_sn}}$, and $P_{\text{amp_n}}$ represent the NEP of the solar background shot noise, signal shot noise, dark current shot noise, and preamplifier noise, respectively, detailed in [96], [98].

2) SINR: In dense or multi-node 3-D UOWC networks, interference from other transmitting nodes can become a dominant performance-limiting factor. In such scenarios, the SINR is a more comprehensive metric than SNR [99]. An analytical approximation for the average SINR in a 3-D network, assuming interference nodes are uniformly distributed in a sphere, has been derived [99]. A full expression for this approximation, incorporating the expected value of interference power and a numerically determined coefficient function $f(c(\lambda), n)$ to improve accuracy, is presented as:

$$\text{SINR} = \frac{\gamma^2 \left[P_T \, \eta_T \eta_R \ \exp\big(-c(\lambda)R\big) \ \frac{A_{\text{R}}}{2\pi R^2}\right]^2}{\left[2e\gamma \, B_n \left(\mathbb{E}\left[\sum_{i=1}^{n} P_{\text{R},i}\right]\right) + 2e \, I_{\text{bg}} I_2 B_n + \sigma_{\text{thermal}}^2 + \gamma^2 \left(\mathbb{E}\left[\sum_{i=1}^{n} P_{\text{R},i}\right]\right)^2\right] f\left(c(\lambda), n\right),} \tag{53}$$

where γ is the Optical-to-Electrical (O/E) conversion efficiency. $P_{\text{R},i}$ is the optical power received at the optical access point from the i-th interfering node [99]. This model provides a crucial tool for analyzing and designing protocols for multi-node UOWC networks.

While physics-based analytical models provide a rigorous foundation for UWOC system design, their predictive accuracy depends strongly on assumptions regarding scattering order, medium homogeneity, temporal dispersion, and receiver geometry. In highly turbid waters, long-range links, or high-data-rate systems, simplifying assumptions such as single-scattering attenuation or reduced-order radiative transfer formulations may no longer hold. A comprehensive discussion of these regime-dependent limitations is provided in Section XI.

D. ML–Based Propagation Models in UOWC

UOWC channels are dominated by wavelength-dependent absorption, strong multiple scattering caused by suspended particles, and turbulence induced by temperature and salinity gradients. These mechanisms produce highly nonlinear and environment-specific propagation behavior, making accurate analytical modeling challenging and motivating data-driven approaches. Table X provides a concise overview of ML-based propagation modeling approaches in UOWC systems, highlighting their learned tasks, input features, and targeted communication objectives. Based on the modeling objective, these ML-based UOWC studies can be grouped into two primary categories:

Joint channel characterization and signal recovery. Within this category, [100] employs supervised learning to implicitly model underwater channel behavior while simultaneously addressing channel classification, estimation, and signal detection tasks. A DNN is trained using synthetic channel realizations generated from established underwater channel formulations across clear, coastal, and turbid water conditions. The model learns the nonlinear mapping between received optical signal features and underlying channel states. Importantly, the authors further validate their approach using experimental measurements obtained from a controlled water tank setup, where turbulence and scattering effects are emulated using air blowers, water droppers, and Maalox antacid to reproduce oceanic-like conditions. The experimental results demonstrate that the trained DNN generalizes to previously unseen channel realizations, indicating that composite propagation effects—such as multipath scattering and thermally induced fluctuations—are implicitly captured by the learned model. Nevertheless, the learning architecture itself does not explicitly embed radiative transfer equations or scattering coefficients; rather, physical modeling is reflected through the data generation and validation process.

Learning-based adaptive transmission control. In a different direction, [101] formulates beamwidth and orientation adaptation in UOWC as an RL problem. Instead of explicitly estimating channel parameters, the transmitter interacts with the underwater propagation environment and updates its transmission policy based on reward signals derived from received power or SNR. In this formulation, the channel is treated

as a stochastic environment, and beam-control strategies are learned through iterative exploration. Similar to the supervised approach above, the learning mechanism does not explicitly incorporate closed-form attenuation or scattering expressions, but instead relies on interaction-driven adaptation.

Across the two categories identified above, ML techniques provide flexibility in handling nonlinear scattering behavior and rapidly varying underwater conditions. The supervised framework in [100] enables implicit channel characterization and joint signal recovery with experimental validation, while the RL-based strategy in [101] supports adaptive transmission control under environmental uncertainty.

While RL has been extensively studied in underwater acoustic communications for adaptive modulation, transmission control, and resource allocation [102]–[105], the underlying propagation physics in acoustic channels differ fundamentally from optical channels in terms of attenuation mechanisms, Doppler characteristics, and multipath structure. Consequently, insights from acoustic-domain learning cannot be directly transferred to UOWC systems, where absorption and scattering dominate signal degradation. However, ML-based UOWC propagation modeling remains inherently data-dependent. Model performance depends strongly on the representativeness of training conditions in terms of turbidity levels, water composition, turbulence statistics, and hardware configurations. Although experimental validation improves confidence in practical applicability, generalization across significantly different underwater environments remains an open challenge. A comprehensive discussion including possible future research direction is provided in Section XI. A summary of ML-based models is shown in Table X.

VII. Underground Propagation Models

A. Propagation Challenges

Accurate characterization of VLC channels in underground mining environments requires consideration of impairments absent in conventional indoor settings. The underground mining environment is one of the most dangerous work scenarios, due to the physical insecurity of the tunnels and the presence of external agents from excavations, such as dust, toxic components, and gases [107]. These conditions necessitate a stable and robust communication system to ensure miners' safety and operational efficiency.

The propagation of optical signals underground is affected by numerous physical phenomena including absorption, reflection, diffraction, and scattering. Key challenges include:

- **Complex Geometries:** Narrow, irregular tunnels with uneven cross-sections and rough, non-planar walls induce complex multipath propagation and cause significant variability in link distance. The workface is often surrounded by metallic hydraulic supports, and the cutter is a movable metallic machine, which creates electrostatic shielding and significant degradation of electromagnetic waves [108].
- **Misalignment:** Uneven mounting surfaces for LEDs and PDs, coupled with tunnel deformations, often lead to their rotation and tilt, further destabilizing the link gain [109].
- **Atmospheric Conditions:** Airborne dust particles, particularly coal dust which can be encapsulated by moisture, scatter and attenuate optical beams, accelerating signal degradation and exacerbating ISI. The size and shape of these particles significantly influence the scattering properties of the environment [108].
- **Shadowing:** Both stationary installations and mobile heavy machinery frequently obstruct the optical path, producing shadowing effects and unpredictable signal fades [109].

These intertwined phenomena compel the development of sophisticated channel-modeling techniques to ensure reliable VLC system design and operation in underground mining scenarios. Fig. 9 presents the geometric link components and environmental impairments inherent to a Single-Input Single-Output (SISO) underground mining (UM) downlink system. The transmitter consists of a randomly positioned and oriented LED mounted between the ceiling and the wall, while the photodiode receiver is located at a height determined by the common placement of devices within the tunnel. Unlike conventional indoor VLC environments, underground mine channels are characterized by irregular tunnel geometries, non-Lambertian reflections from rough coal surfaces, and severe particulate scattering caused by coal dust. As a result, channel models developed for indoor VLC cannot be directly applied without significant modification.

B. Analytical Propagation Models

Analytical models for VLC channels in underground mines are based on the principles of optical wave propagation, considering both LoS and NLoS components.

1) Lambertian Emission and DC Channel Gain: The radiation pattern of LEDs follows the generalized Lambertian model, where the radiant intensity is given by (28), and the Lambertian order m is defined in (27).

For a simple LoS link in underground environments, the DC channel gain follows the standard Lambertian model given in (30). This formulation remains applicable, as it primarily depends on the transmitter radiation pattern and receiver geometry rather than the specific propagation environment.

2) Received Optical Power: The received optical power is the sum of the LoS component and the NLoS components arising from reflections. When a direct LoS path exists, the received power component follows the same formulation as in indoor optical channels (31). However, due to frequent blockage and strong scattering in underground environments, the overall received power is typically influenced by both LoS and multiple NLoS components.

In underground mines, the NLoS component can become dominant in scenarios involving tunnel bends, machinery-induced blockage, or misalignment, making accurate modeling of reflected paths essential.

Reflections from tunnel walls, which are often irregular and made of materials with low reflectivity like coal, contribute to the NLoS component. The non-LoS DC channel gain from

TABLE X: Summary of ML-based propagation modeling approaches in UOWC systems.

Model	Environment	Learned Relationship / Task	Key Inputs	Focus
[100]	Underwater (simulated UOWC: clear/coastal/turbid)	CNN-based joint learning: channel classification + channel estimation + signal detection (supervised, synthetic labels)	Synthetic channel realizations/ received-signal features	End-to-end learning for UOWC channel/detection
[101]	Underwater (simulated seawater/ ocean/coastal/harbor)	RL for beamwidth/orientation adaptation to improve received power/SNR	Simulated channel feedback/reward signals	Adaptive beam control in UOWC

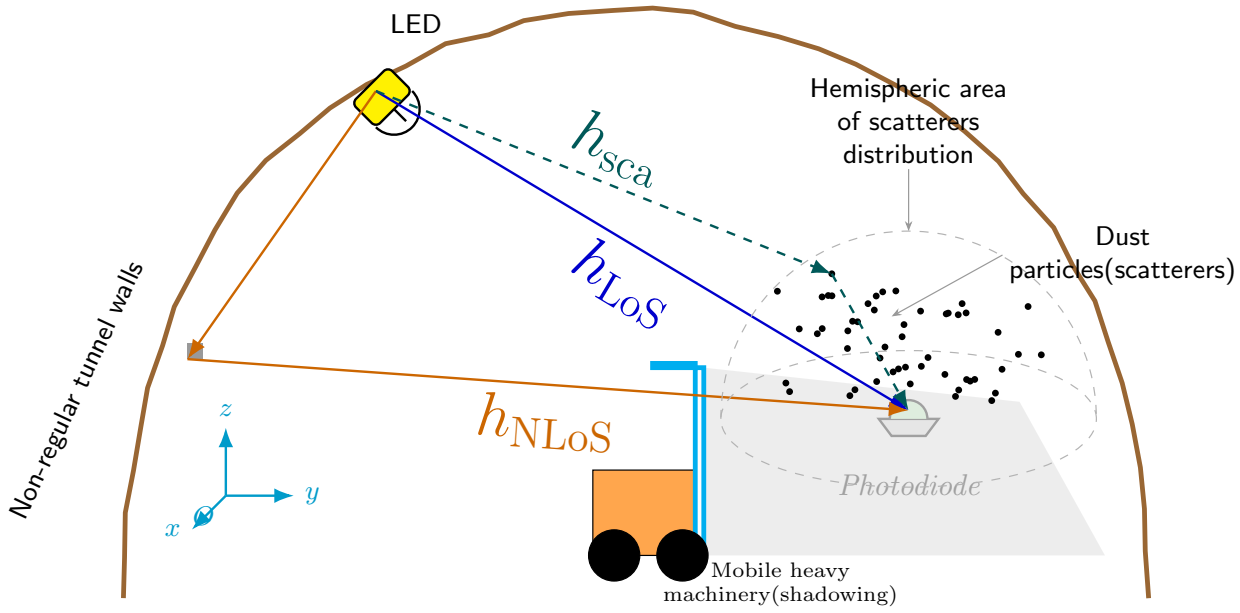


Fig. 9: Typical geometric configuration of Underground Mining VLC (UM-VLC) system, illustrating the propagation paths and environmental impairments [106].

reflections can be modeled by dividing the wall surfaces into small reflector elements and summing their contributions. For a single reflection, the channel gain is given by [20]:

$$H_{\mathrm{NLoS}} = \int_{\mathrm{walls}} \frac{A_R(m+1)}{2\pi^2 R_1^2 R_2^2} \rho_k \cos^m(\phi)\cos(\alpha)\cos(\beta) \times T_s(\omega) g(\omega)\cos(\omega) dS_{\mathrm{wall}}, \tag{54}$$

where R_1 is the distance from the LED to the reflection point (m), R_2 is the distance from the reflection point to the PD (m), ρ_k is the surface reflectivity, ϕ is the angle of irradiance to the reflection point (rad), α is the angle of incidence on the reflection point (rad), and β is the angle of irradiance from the reflection point (rad), where dS_{wall} denotes an infinitesimal surface element of the reflecting wall (m^2).

In addition to direct and wall-reflected components, suspended dust particles can introduce a scattering contribution to the received signal. Therefore, the total received optical power can be written as

$$P_{\mathrm{R,total}} = P_{\mathrm{R,LoS}} + P_{\mathrm{R,NLoS}} + P_{\mathrm{R,scat}}, \tag{55}$$

where $P_{\mathrm{R,scat}}$ (W) captures the dust-induced scattered component, which can be modeled statistically (e.g., using a 3D/hemispherical particle distribution around the receiver) [106]. In general, it depends on dust concentration, particle size distribution, and the scattering geometry around the receiver.

3) Coal Workface Channel Model: A specific model for a fully mechanized coal workface was proposed by Zhai and Zhang [108]. This environment is divided into two communication spaces by hydraulic support columns. The model calculates the energy coupling of LoS signals between these two spaces, considering the blocking effect of the metal columns. The coupling space angle γ_n (rad) for the n-th space between columns is given by [108]:

$$\gamma_n = \beta_n - \beta_{n-1} - \theta_n - \theta_{n-1}, \quad n \geq 2 \tag{56}$$

$$\gamma_1 = 2(\beta_1 - \theta_1), \tag{57}$$

where β_n (rad) is the angle from the transmitter to the n-th column center and θ_n (rad) is the angle subtended by the column radius from the transmitter's perspective. This model helps in finding an optimized transmitter location to maximize coupled energy.

4) CIR: The optical channel can be characterized by its impulse response, $h(t)$, which includes all signal paths. The

general optical channel model is given by [20]:

$$y(t) = R_s\big[x(t) \otimes h(t)\big] + n(t), \tag{58}$$

where $y(t)$ is the received signal, $n(t)$ is the AWGN.

For a LoS link, the CIR can be modeled as a delayed Dirac delta function:

$$h_{\mathrm{LoS}}(t) = H_{\mathrm{LoS}}(0) \cdot \delta(t - d/c) \tag{59}$$

where c is the speed of light (ms^{-1}). The NLoS component, resulting from multiple reflections, introduces multipath dispersion, which can be modeled by summing the contributions from all reflection paths.

5) Noise and SNR: The total noise in the system, $\sigma^2_{\mathrm{total}}$, consists of shot noise and thermal noise. The shot noise variance is given by:

$$\sigma^2_{\mathrm{shot}} = 2eR_s(P_{\mathrm{R,total}} + P_{\mathrm{amb}})G_N \tag{60}$$

where P_{amb} is the ambient optical power (W). Ambient light from sources other than the transmitter is a significant noise component.

The thermal noise variance from the amplifier is:

$$\sigma^2_{\mathrm{thermal}} = I_N^2 G_B \tag{61}$$

where I_N is the input current noise density ($\mathrm{A}/\sqrt{\mathrm{Hz}}$) and G_B is the amplifier bandwidth (Hz).

The total electrical SNR is then [109]:

$$\begin{aligned}\mathrm{SNR} &= \frac{(R_s P_{\mathrm{R,total}})^2}{\sigma^2_{\mathrm{thermal}} + \sigma^2_{\mathrm{shot}}} \\ &= \frac{(R_s P_{\mathrm{R,total}})^2}{I_N^2 G_B + 2eR_s(P_{\mathrm{R,total}} + P_{\mathrm{amb}})G_N}\end{aligned} \tag{62}$$

While analytical models provide valuable insight into underground optical propagation, their validity depends strongly on geometric regularity, particulate stability, and link continuity assumptions. In practical subterranean environments, irregular tunnel structures, dynamic dust concentrations, stochastic obstructions, and mechanical misalignment may violate these assumptions, reducing predictive accuracy. A detailed discussion of these regime-dependent limitations and their implications for system design is provided in Section XI.

C. ML–Based Propagation Models

While ML-based modeling for underground optical communication remains relatively nascent, existing works can be interpreted within a broader data-driven modeling framework.

Time-series channel modeling using learning-based system identification. In underground environments, propagation characteristics exhibit strong temporal variability due to dynamic factors such as dust movement, airflow, and machinery activity. As a result, some studies formulate channel modeling as a nonlinear time-series prediction problem. For example, [110] employs a nonlinear autoregressive model with exogenous inputs (NARX) implemented using a neural network architecture, expressed as

$$y(t) = F\big(y(t-1), \ldots, y(t-n_b), u(t-1), \ldots, u(t-n_a)\big) + e(t), \tag{63}$$

where $F(\cdot)$ is a nonlinear function approximated by the neural network, $u(t)$ denotes the system input signals, n_a and n_b represent the input and output model orders, and $e(t)$ is a white-noise term capturing unmodeled disturbances. Such models capture the temporal dependencies of the channel by combining past outputs and inputs, enabling prediction of future channel behavior.

Learning nonlinear channel dynamics from measurements. Unlike analytical models that assume predefined attenuation or scattering structures, ML-based approaches learn the channel response directly from measured data. In [110], the nonlinear function is approximated using a two-layer MLP trained via the Levenberg–Marquardt algorithm, achieving prediction errors within ± 0.2 V under experimental underground conditions. This demonstrates the ability of neural networks to capture complex nonlinearities inherent in underground VLC channels.

Implications for underground propagation modeling. Although current literature is limited, these approaches highlight the potential of ML for environments where channel behavior is difficult to model analytically. Underground optical channels are influenced by irregular tunnel geometries, varying dust concentrations, and dynamic obstructions, making purely physics-based models less tractable. Data-driven models can therefore provide flexible approximations of channel behavior, particularly for prediction and monitoring tasks.

Across these perspectives, ML-based propagation modeling in underground environments remains in an early stage compared to air or indoor scenarios. While initial results demonstrate strong capability in capturing nonlinear temporal dynamics, broader validation across diverse tunnel conditions and integration with physics-based insights will be essential to ensure robustness and generalization.

VIII. Deep Space Propagation Models

Deep-space optical communication (DSOC) relies on the transmission of narrow, high-precision laser beams across vast interplanetary distances [111]. With the growing deployment of low Earth orbit (LEO) satellite constellations, space based optical communication in the form of optical inter satellite links (OISLs) and ground to space (G2S) links is gaining momentum. Unlike RF systems, these optical links operate with extremely small beam divergences, making the communication channel highly sensitive to pointing accuracy, spacecraft vibration, and alignment errors. The optical signal can also be weakened by photon scarcity over long distances. As a result, maintaining stable and reliable optical links in deep space requires precise beam control, efficient photon detection, and robust mitigation of noise and attenuation effects. A typical DSOC scenario is portrayed in Fig. 10.

A. Deep Space Communications Challenges

- Ultra-long communication distances exceeding millions of kilometers, causing extreme free-space path loss [22].
- Interference from other missions, planetary bodies, and solar radiation [112].

- Solar scintillation, atmospheric scintillation, and other channel impairments [113].
- High sensitivity to Sun-Earth-probe geometry and line-of-sight blocking.
- Size, weight, and power (SWaP) constraints of spacecraft systems.
- Long one-way latency causing delays in control and communication [114].
- Requirements for high transmitter power and large receiver apertures [115].
- Limited RF bandwidth and susceptibility to eavesdropping [116].

B. DSOC Challenges

- Extremely tight beam pointing accuracy requirements (micro-radian level).
- Weak received optical power due to long distances and narrow beams.
- Atmospheric losses including scattering, absorption, turbulence, and cloud blockage at ground receivers.
- Need for large-aperture optical telescopes and precise tracking mechanisms.
- Sensitivity to pointing jitter and spacecraft attitude disturbances.
- Complexity of photon-counting receivers and modulation schemes [117].
- SWaP constraints for high-efficiency lasers and optical terminals.
- Reliability issues due to weather-dependent ground station availability.

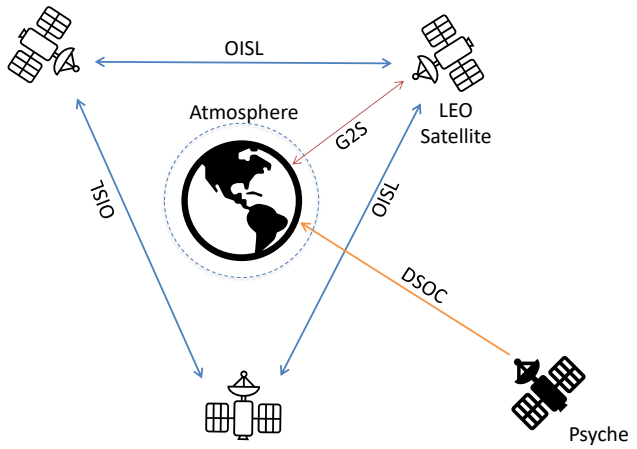


Fig. 10: Conceptual illustration of DSOC scenarios.

C. Analytical Modeling

In DSOC, the received power is determined by geometric free-space loss, the transmitter and receiver aperture gains, pointing accuracy, and the optical efficiencies of the terminals. Following the link-budget formulation in [118], the received power after the receiver telescope and before the photon detector is expressed as:

$$P_{R_{\mathrm{ap}}} = P_T\, G_t\, G_r\, L_{\mathrm{fs}}\, h_{\mathrm{atm}}\, L_c\, L_s\, h_{\mathrm{pe}}\, \eta_T\, \eta_R, \tag{64}$$

where G_t and G_r are the transmitter and receiver aperture gains, respectively, L_{fs} is the free-space loss factor, L_c is the cirrus-cloud transmittance factor, L_s is the scintillation loss factor,

Assuming a diffraction-limited transmitter with no obscuration, the transmitter gain is [119]:

$$G_t = \frac{32}{\theta^2}, \tag{65}$$

When the obscuration ratio is $\gamma_r = b_r/D_r$, where b_r is the secondary mirror diameter (m), the receiver gain is [120]:

$$G_r = \left(\frac{\pi D_r}{\lambda}\right)^2 \left(1-\gamma_r^2\right). \tag{66}$$

For ground-to-space and near-Earth deep-space links, atmospheric turbulence introduces random irradiance fluctuations that cannot be fully captured by a single scintillation loss factor. Several works model turbulence using the refractive-index structure parameter C_n^2, often described by the Hufnagel–Valley model as a function of altitude and wind speed [113]. For spherical-wave propagation, the resulting irradiance scintillation can be quantified using the Rytov variance σ_R^2, which serves as a measure of turbulence strength. The received optical power may then be modeled as a random process [121]. This statistical description enables outage probability and error-rate analysis under turbulence-induced fading.

The free-space loss factor accounts for geometric spreading over the link distance R (m) and is given by the Friis transmission formula [122]:

$$L_{\mathrm{fs}} = \left(\frac{\lambda}{4\pi R}\right)^2. \tag{67}$$

Beyond the scalar free-space loss factor in (67), deep-space optical propagation can be modeled using Gaussian-beam wave optics, in which the optical field expands with distance according to the beam radius w_z. For a diffraction-limited Gaussian beam, the transverse irradiance distribution is given by [123]:

$$I = I_0 \left(\frac{w_0}{w_z}\right)^2 \exp\!\left(-\frac{2r^2}{w_z^2}\right), \tag{68}$$

where I_o is the peak irradiance and r (m) is the radial distance in the receiver plane where the intensity is measured. The received power is then obtained by integrating (68) over the receiver aperture, naturally capturing beam spreading, aperture truncation, and off-axis misalignment [23]. This formulation is equivalent to the Friis-based expression when evaluated under on-axis, far-field conditions, but provides additional insight into spatial coupling effects.

The atmospheric transmittance factor L_a models absorption and scattering losses and is commonly expressed using the Beer–Lambert law as $L_a = \exp(-\tau_a \sec\theta_e)$ [124] [125], where τ_a denotes the vertical atmospheric optical depth and θ_e is the elevation angle of the optical link. The cirrus-cloud transmittance L_c accounts for attenuation due to thin clouds and is often modeled as $L_c = \exp(-\tau_c)$ [124], where τ_c represents the cirrus cloud optical depth. Scintillation loss L_s captures turbulence-induced irradiance fluctuations; under

weak turbulence, the mean loss can be approximated as $L_s = \exp(-\sigma_R^2/2)$ [126].

Pointing loss arises from boresight error and spacecraft jitter. For a given total pointing-error variance σ_p^2 and probability level p_0, the pointing-loss factor can be approximated as [127]:

$$h_{\mathrm{pe}} = p_0^{\left(\frac{\theta^2}{16\sigma_p^2}\right)}. \tag{69}$$

Pointing loss in deep-space optical links is often modeled deterministically using (69); however, several studies treat pointing error as a random process arising from spacecraft attitude jitter and tracking inaccuracies. When azimuth and elevation errors are modeled as independent zero-mean Gaussian variables, the resulting radial pointing error follows a Rayleigh distribution, while biased tracking errors lead to a Rician distribution [22]. The instantaneous pointing loss can then be expressed as a function of pointing errors for Gaussian beams, or via aperture-pattern expressions involving Bessel functions for circular apertures. This formulation allows the pointing channel to be treated as a fading process and supports outage-based performance metrics in addition to average received power.

Finally, after the optical signal passes through the receiver optics and is incident on the photon-counting detector, the detected signal power is [128]:

$$P_R = P_{R_{\mathrm{ap}}}\, \eta_{\mathrm{det}}\, L_b\, L_j, \tag{70}$$

where η_{det} is the detector quantum efficiency, and L_b and L_j are the blocking and jitter loss factors, respectively.

D. ML-based Modeling

ML has not yet been widely incorporated into operational DSOC systems, which remain dominated by analytically designed photon-counting receivers and coded PPM detection architectures [129]. State-of-the-art DSOC receiver designs, such as the photon-counting array architecture presented by Mendenhall et al. [117], describe deterministic signal processing pipelines optimized for extreme photon scarcity, long propagation distances, and background-limited operation. These architectures define the practical baseline required for integration in learning-based enhancement.

Results from several related optical communication domains provide clear evidence of the potential value of ML-based modeling. Deep learning has been shown to improve optical signal detection under complex, non-Gaussian impairments. Bart et al. demonstrated a CNN detector that outperforms classical thresholding for turbulence-impaired optical links using experimentally collected intensity data [130]. Similarly, Lohani et al. used generative ML models to compensate severe optical distortions and restore reliable communication through scattering and turbulence [131]. These approaches illustrate that data-driven receivers can learn channel statistics that are difficult to model analytically, which is relevant to DSOC scenarios involving pointing jitter, background noise fluctuations, and detector nonidealities.

Neural models have also been used successfully in photon-counting front ends. Studies shows that a radial basis function neural network can learn the nonlinear detection characteristics of a silicon photomultiplier and significantly improve symbol classification in a photon-limited link [132]. While developed for VLC, the ability to learn decision boundaries directly from photon-counting statistics is closely aligned with the needs of DSOC receivers that rely on single-photon avalanche diode or superconducting nanowire detectors.

Together, these studies indicate that ML may offer pathways for enhanced demodulation, compensation of pointing-induced fading, and improved photon-limited detection in future DSOC systems.

IX. Summary of Received-Power and Propagation Models

To facilitate cross-environment comparison and improve the accessibility of the surveyed literature, this section consolidates the principal received-power, channel-gain, and signal-to-noise expressions used across OWC environments. While the preceding sections discuss each model within its physical and methodological context, the Tables XI-XV below provide a compact, equation-level reference spanning atmospheric, indoor VLC/IR, underwater, underground, and deep-space optical links.

The goal of this consolidation is not to introduce new models, but to offer a unified mathematical snapshot that highlights common structural elements, environment-specific impairments, and recurring modeling assumptions. This summary is intended to support readers in quickly identifying suitable models for system design, performance analysis, and optimization across diverse optical propagation scenarios.

By presenting these equations side by side, the survey emphasizes both the shared mathematical foundations of optical wireless channels and the distinct environment-specific phenomena that necessitate specialized modeling approaches.

X. Optical Wireless Communication Model (OWCM) - Open source python Library

work in progress - zaheen !!

Despite the growing interest in OWC, including FSO, VLC, and UOWC, very few open-source software libraries exist for comprehensive channel modeling in these domains. In practice, most published studies rely on ad-hoc MATLAB scripts or isolated simulation implementations, which limits reproducibility, extensibility, and cross-comparison of results. One of the few available open-source frameworks is gr-owc, developed by the Ubiquitous Communications and Networking Lab (UCaN) at the University of Massachusetts Boston, which extends GNU Radio with OWC blocks for channel modeling and modulation. The framework includes components such as Lambertian transmitters, photodetector models, optical filters, and modulation blocks (e.g., OOK, PAM, and PPM), but it is primarily designed for signal-processing experimentation within the GNU Radio ecosystem and focuses mainly on indoor LoS channels using Lambertian propagation models [133]. Another related tool is Py6S, a Python interface to the Second Simulation of a Satellite Signal in the Solar Spectrum (6S) radiative transfer model, which is widely used

TABLE XI: Atmospheric OWC channel equations.

Purpose	Equation	Ref.
Unified multiplicative channel model	$P_R = P_T h_{\text{geo}} h_{\text{atm}} h_{\text{turb}} h_{\text{pe}}$	(4)
Classical FSO range equation	$P_R = P_T \left[\frac{A_r}{(D_T + R\theta)^2}\right] T K e^{-\alpha_{atm} R}$	(5)
Geometric loss (aperture-based)	$h_{\text{geo}} = \left(\frac{D_R}{D_T + R\,\theta}\right)^2$	(6)
Gaussian-beam geometric coupling	$h_{\text{geo}} = A_0 = \left[\text{erf}\left(\frac{\sqrt{2}\,a}{w_z}\right)\right]^2$	(7)
Exponential approximation of coupling	$A_0 \simeq 1 - \exp\left(-\frac{2a^2}{w_z^2}\right)$	(8)
Beer–Lambert transmittance	$h_{\text{atm}} = e^{-\alpha_{\text{atm}} R}$	(9)
Visibility-based attenuation (Kim)	$\alpha_{\text{atm}} = \frac{3.91}{V}\left(\frac{\lambda}{550}\right)^{-q}$	(10)
Rytov variance	$\sigma_R^2 = 1.23 C_n^2 k^{7/6} R^{11/6}$	(13)
Gamma–Gamma irradiance PDF	$f(h_{turb}) = \frac{2(\alpha\beta)^{(\alpha+\beta)/2}}{\Gamma(\alpha)\Gamma(\beta)} h_{turb}^{\frac{\alpha+\beta}{2}-1} K_{\alpha-\beta}\left(2\sqrt{\alpha\beta\, h_{turb}}\right)$	(14)
Rayleigh pointing-error PDF	$f_r(r) = \frac{r}{\sigma_s^2} e^{-r^2/(2\sigma_s^2)}, \qquad r \geq 0.$	(17)
Hoyt (Nakagami-q) pointing-error PDF	$f_r(r) = \frac{r}{q\,\sigma_y^2} \exp\left(-\frac{r^2(1+q^2)}{4q^2\sigma_y^2}\right) I_0\left(\frac{r^2(1-q^2)}{4q^2\sigma_y^2}\right)$	(18)
Pointing-error loss	$h_{\text{pe}}(r) = \exp\left(-\frac{2r^2}{w_{z,\text{eq}}^2}\right)$	(21)
Rician distribution pdf	$f_r(r) = \frac{r}{\sigma_s^2} \exp\left(-\frac{r^2+s^2}{2\sigma_s^2}\right) I_0\left(\frac{rs}{\sigma_s^2}\right), \qquad r \geq 0$	(19)
Beckmann radial PDF	$f_r(r) = \frac{r}{2\pi\sigma_x\sigma_y} \int_0^{2\pi} \exp\left(- \frac{(r\cos\theta - \mu_x)^2}{2\sigma_x^2} - \frac{(r\sin\theta - \mu_y)^2}{2\sigma_y^2}\right) d\theta,$ $r \geq 0$	(20)
SINR (single link)	$\text{SINR}_A = \frac{P_T \tau(R) \lambda^2 A_r}{8\pi^2 d^2 (1-\cos\theta)(N_T + \Gamma * sA)}$	(24)
SINR with pointing & vibration	See Eq. (25)	(25)
SINR (multi-element FSO)	See Eq. (26)	(26)

TABLE XII: Indoor VLC / IR-OWC channel equations.

Purpose	Equation	Ref.
Lambertian order	$m = -\frac{\ln 2}{\ln(\cos\phi_{1/2})}$	(27)
Lambertian radiant intensity	$R_o(\phi) = \frac{m+1}{2\pi} \cos^m(\phi)$	(28)
Optical concentrator gain	$g(\omega) = \frac{n^2}{\sin^2\omega_c}$	(29)
LoS DC channel gain	$H_{\text{LoS}}(0) = \frac{A_d}{d_o^2} R_o(\phi_0) \cos(\omega_0) T_s(\omega_0) g(\omega_0)$	(30)
Received LoS power	$P_{\text{R,LoS}} = P_T H_{\text{LoS}}(0)$	(31)
NLoS impulse response (general)	$h_{\text{NLoS}}(t) = \sum_{k=0}^{K} h_{\text{NLoS}}^{(k)}(t)$	(32)
Single-reflection DC gain	See Eq. (36)	(36)
Total received power	$P_R = P_T \left[H_{\text{LoS}}(0) + \int dH_{\text{NLoS}}^1(0)\right]$	(37)
Total noise variance	$\sigma_{\text{total}}^2 = \sigma_{\text{thermal}}^2 + \sigma_{\text{shot}}^2$	(38)
SNR (VLC)	$\text{SNR} = \frac{(R_s P_R)^2}{\sigma_{\text{total}}^2}$	(41)

in atmospheric remote sensing to estimate absorption and scattering effects at optical wavelengths. While Py6S can model atmospheric transmittance, it does not capture critical communication-specific phenomena such as turbulence-induced fading, pointing errors, receiver FoV constraints, or link geometry, and therefore cannot directly support end-to-end optical wireless communication simulations [134]. Consequently, there remains a clear need for a modular and extensible software framework capable of modeling diverse optical wireless channel environments.

To address this gap, we introduce the OWC Models (OWCM) library, a Python-based framework designed to provide a unified and extensible platform for modeling OWC channels. OWCM defines a standardized abstraction for optical links through structured representations of transmitters, receivers, environments, and propagation models.

A. Library Architecture

The OWCM library follows a five-layer architecture, illustrated in Fig. 11, that strictly separates user-facing con-

TABLE XIII: Underwater OWC channel equations.

Purpose	Equation	Ref.
Extinction coefficient	$c(\lambda) = a(\lambda) + b(\lambda)$	(45)
Beer–Lambert underwater loss	$L_{\text{pr}} = \exp[-c(\lambda)z]$	(46)
LoS received power	$P_{R,LoS} = P_T \eta_T \eta_R L_{pr}\left(\lambda, \frac{d}{\cos\theta}\right) \frac{A_{Rec}\cos\theta}{2\pi d^2(1-\cos\theta_0)}$	(47)
Retroreflector link power	$P_{R,Retro} = P_T\, \eta_T \eta_R \eta_{Retro}\, L_{pr}\left(\lambda, \frac{2d}{\cos\theta}\right) \times \left[\frac{A_{Retro}\cos\theta}{2\pi d^2(1-\cos\theta_0)}\right]\left[\frac{A_{Rec}\cos\theta}{\pi(d\tan\theta_{0retro})^2}\right]$	(48)
Reflective (NLoS) link	$P_{R,\text{ref}} \approx A_{\text{rec}} f_{R,\text{ref}}(\phi)$	(49)
Auxilary function of NLoS link	$f_{R,\text{ref}}(\phi) = \begin{cases} \frac{P_T\eta_T\eta_R L_{\text{pr}}(\lambda, h+x)\cos\phi}{2\pi(h+x)^3(\cos\theta_{\min}-\cos\theta_{\max})} \times \left[\frac{1}{2}\left(\frac{\tan(\theta_t-\phi)}{\tan(\theta_t+\phi)}\right)^2 + \left(\frac{\sin(\phi-\theta_t)}{\sin(\phi+\theta_t)}\right)^2\right], \\ \theta_{\min} \le \phi \le \theta_c, \\ \frac{P_T\eta_T\eta_R L_{\text{pr}}(\lambda, h+x)\cos\phi}{2\pi(h+x)^3(\cos\theta_{\min}-\cos\theta_{\max})}, \\ \theta_c < \phi \le \theta_{\max}. \end{cases}$	(50)
Electrical SNR	See Eq. (51)	(51)
Average SINR (3D network)	See Eq. (53)	(53)

TABLE XIV: Underground OWC channel equations.

Purpose	Equation	Ref.
NLoS wall-reflection gain	See Eq. (54)	(54)
Total received power	$P_{\text{R,tot}} = P_{\text{R,LoS}} + P_{\text{R,NLoS}} + P_{\text{R,scat}}$	(55)
Optical channel model	$y(t) = R_s\left[x(t) \otimes h(t)\right] + n(t)$	(58)
Electrical SNR	See Eq. (62)	(62)

TABLE XV: Deep space OWC channel equations.

Purpose	Equation	Ref.
Link-budget formulation	$P_{r,\text{ap}} = P_t G_t G_r L_{fs} L_a L_s L_{pt} \eta_t \eta_r$	(64)
Transmitter gain	$G_t = \frac{8}{w_{0,\text{rad}}^2}$	(65)
Receiver gain	$G_r = \left(\frac{\pi D_r}{\lambda}\right)^2 (1-\gamma_r^2)$	(66)
Free-space loss	$L_{fs} = \left(\frac{\lambda}{4\pi R}\right)^2$	(67)
Pointing loss	$L_{\text{pt}} = \exp\left(-\frac{2\sigma_p^2}{\theta_{1/e^2}^2}\right)$	(69)

figuration, physical modeling, and model composition. At the top, the *User API Layer* exposes a fluent builder interface: `Channel.environment(`*`name`*`)` returns an `EnvironmentBuilder`, to which `.defaults(`*`recipe`*`)` applies a named model stack, `.model(`*`category`*`, `*`id`*`)` overrides individual components, and `.build(**kwargs)` instantiates a `ConfiguredChannel` encapsulating a `Link`, an `Environment`, and a `CompositeChannel`. The *Core Engine* underpinning this interface consists of four subsystems: a decorator-driven model registry indexed by a three part key (environment, category, model_id); a composite channel that accumulates sub-model contributions; a named recipe store mapping scenario identifiers (e.g., `mvp`, `clear_air`, `dense_fog`) to category–model dictionaries; and shared type and random-number-generation primitives (`Link`, `Environment`, `get_rng`).

Five *Environment Modules*—`atmospheric`, `indoor`, `underwater`, `underground`, and `space`—each provide a `models/` directory of physics implementations and a `recipes.py` file; importing a package is sufficient to populate the registry. Each model conforms to a common `ChannelModel` protocol that mandates three methods: `gain(link, env)`, returning a deterministic scalar gain; `sample(link, env, n)`, returning an n-element `numpy` array of stochastic realizations; and `describe()`, returning a JSON-serializable metadata dictionary.

Finally, the *Channel Output Interface* of `ConfiguredChannel` exposes `received_power()`, `outage(`*`threshold`*`, n)`, `gain()`, and `describe()`, thereby supporting deterministic link-budget evaluation, Monte Carlo outage studies, and structured logging for reproducibility within a single unified object.

OWCM is governed by three principles. Models are organized according to the OWC survey taxonomy as shown

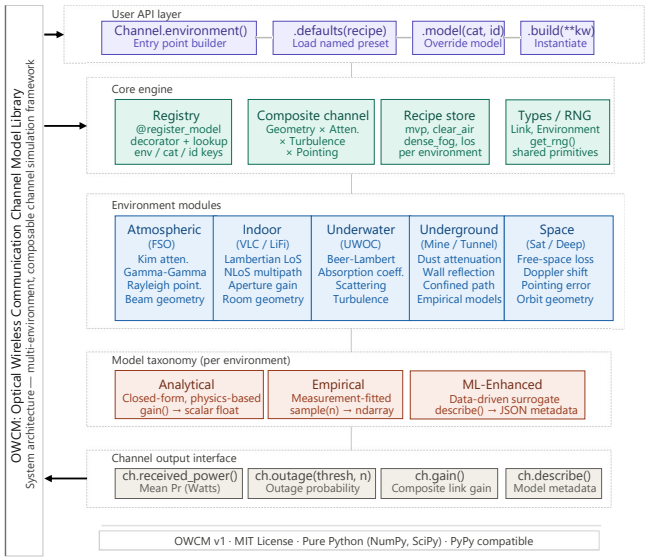


Fig. 11: OWCM system architecture

in this paper, mapping each class to an identifiable propagation effect so that implementations can be validated against published expressions and replaced independently. The `CompositeChannel` implements the standard link-budget factorization $h_{\text{total}} = \prod_i h_i$, keeping model classes small and stateless, while additive physics (e.g., LoS plus NLoS in indoor VLC) is encapsulated within a single registered model. Finally, the `gain`/`sample` protocol enforces a clean separation between scalar mean-channel evaluation and stochastic sampling, preventing accidental substitution of one for the other. These principles yield uniform API consistency across all five propagation domains, low-friction extensibility—new models require only a decorated class and an optional recipe entry—and full parameter traceability through `describe()`, which carries equation references suitable for embedding directly in reproducibility logs or supplementary material.

XI. Open Challenges, and Research Directions

Optical wireless propagation models provide essential tools for system design and performance analysis. However, their predictive accuracy depends on the validity of simplifying assumptions regarding environmental conditions, propagation physics, and transmitter–receiver geometry. In practice, optical wireless links operate in diverse environments, including atmospheric, underwater, underground, space, and indoor settings, where each characterized by distinct physical impairments and channel dynamics. Depending on the environment, these impairments may include absorption, scattering, turbulence, obstruction, background radiation, geometric shadowing, and transmitter–receiver misalignment. As a result, many analytical propagation models are valid only within specific operating conditions and may lose predictive accuracy when applied outside their intended regimes. This section summarizes the major challenges of physics-based propagation models across different optical wireless environments and highlights emerging research directions toward more robust modeling frameworks, including hybrid analytical–empirical approaches and data-driven techniques.

A. Challenges of Analytical Atmospheric Models

Analytical models provide valuable insight into atmospheric optical propagation. However, their predictive accuracy depends on simplifying assumptions regarding turbulence strength, scattering mechanisms, and link geometry. When these assumptions are violated, several modeling challenges arise.

Saturation of Scintillation. The Rytov approximation, commonly used to characterize atmospheric turbulence, assumes weak-to-moderate turbulence conditions and forms the basis for defining the Rytov variance σ_R^2. Under this assumption, intensity fluctuations increase monotonically with the refractive index structure parameter C_n^2. However, for strong turbulence regimes or long propagation distances, this relationship no longer holds. In such cases, scintillation approaches a saturation regime due to effects such as beam breakup, inner- and outer-scale turbulence interactions, and multiple phase perturbations. Consequently, the widely used log-normal fading model becomes less accurate, necessitating the adoption of more general statistical models such as the Gamma–Gamma or Negative Exponential distributions to capture the observed irradiance statistics.

Multiple Scattering Effects Beyond the Beer–Lambert Law. Atmospheric attenuation is often modeled using the Beer–Lambert law, $h_{\text{atm}} = e^{-\alpha_{atm} R}$, which assumes a single-scattering process in which photons are either absorbed or scattered out of the receiver aperture. This assumption is reasonable for light haze or moderate rain conditions. However, in dense fog or heavy precipitation environments with visibility $V < 0.5$ km, multiple scattering events become significant. Under such conditions, a fraction of forward-scattered photons may re-enter the receiver field of view, depending on the optical configuration. Moreover, multiple scattering introduces temporal dispersion that can cause pulse broadening and potential ISI in high-data-rate systems. As a result, the Beer–Lambert model may underestimate received power and fail to capture delay spread effects in highly scattering environments.

Instability of Visibility-Dependent Wavelength Exponents. Empirical attenuation models, such as the Kim and Kruse models, incorporate a visibility-dependent exponent q to account for wavelength-dependent scattering behavior. While effective for general haze and mist conditions, these models become unreliable in dense fog regimes. When particle sizes become comparable to the optical wavelength, scattering transitions toward the Mie regime, reducing wavelength dependence. Consequently, the commonly assumed attenuation advantage of longer wavelengths (e.g., 1550 nm over 850 nm) may be significantly diminished, limiting the applicability of visibility-based empirical models under severe fog conditions.

Pointing Error and Misalignment Modeling. Many analytical FSO propagation models assume perfect alignment between the transmitter and receiver optical axes. In practical deployments, however, mechanical vibrations, building sway,

wind loading, and platform mobility can introduce pointing errors that cause beam displacement at the receiver plane. These misalignment effects may significantly reduce the received optical power and increase the probability of link outage, particularly for narrow optical beams and long propagation distances. While several statistical models have been proposed to characterize pointing error effects, many analytical formulations approximate the beam displacement at the receiver plane using simplified distributions such as Rayleigh, Beckmann, or Hoyt models and assume stationary alignment conditions. In realistic scenarios, however, pointing errors may exhibit time-varying and correlated behavior due to platform dynamics and environmental disturbances, making accurate modeling of misalignment effects an ongoing challenge in FSO channel modeling.

The challenges outlined above highlight that no single physics-based atmospheric model is universally applicable across all operating regimes. Accurate performance prediction for atmospheric FSO systems requires careful selection of models based on turbulence strength, weather severity, wavelength, and receiver geometry, and, where possible, validation through empirical measurements.

B. Challenges of Analytical Indoor Propagation Models

Oversimplification of Geometric Assumptions. Most analytical models consider idealized room geometry with fixed transmitter-receiver positioning, and simplified Lambertian assumptions. In reality, indoor propagation environment is heavily influenced by room size, receiver location, and reflective material properties. This often results in failure of closed-form models to capture the spatial variability of CIR and receiver power in practical setup. Such limitation eventually pushes to recursive and ray-tracing approaches for high-precision indoor VLC characterization.

Gaps in Modeling of Multipath Dispersion. Although indoor optical channels suffer from multipath dispersion, analytical DC-gain models focuses more on first-order power approximation. By taking into account single reflection-based propagation, such models often neglect the dispersive links where multiple reflected links can arrive with varied delays. As bit rate increases, such simplification starts hurting as ISI becomes dominant factor.

Poor Robustness to Blockage, Shadowing and Mobility. In practical indoor setting, blockage and shadowing from obstructions, device orientation, user mobility can frequently disrupt the LoS and NLoS link geometry, causing time-varying channel behavior and abrupt changes in SNR. Such dynamic behavior often becomes difficult to model through static analytical expressions, resulting in overestimation of link reliability and sudden SNR drop. This motivates the use statistical variability-aware modeling approaches.

C. Challenges in Analytical Underwater Optical Propagation Models

Breakdown of Single-Scattering and Beer–Lambert Models.

The Beer–Lambert law is widely used as a first-order analytical model for underwater optical attenuation, describing signal decay as an exponential function governed by the absorption and scattering coefficients of the medium [97]. This formulation implicitly assumes a single-scattering process in which photons scattered away from the LoS do not contribute to the received signal. While this assumption is reasonable in clear oceanic waters and short-range links, it becomes inaccurate in coastal or harbor environments characterized by high turbidity. In such conditions, multiple scattering events dominate, allowing a fraction of photons to reach the receiver via indirect paths. Consequently, Beer–Lambert-based models tend to underestimate received power and fail to capture the spatial and temporal dispersion introduced by volumetric scattering.

Limitations of Radiative Transfer Simplifications. More advanced UWOC models are derived from the radiative transfer equation (RTE), which provides a comprehensive description of light propagation in scattering media. However, closed-form or reduced-order solutions to the RTE often rely on assumptions such as homogeneous water properties, steady-state illumination, or simplified phase functions. These assumptions may not hold in dynamic underwater environments where particle concentration, salinity, and temperature vary spatially and temporally. As a result, RTE-based analytical approximations may lose accuracy when applied beyond their intended environmental conditions.

Neglect of Temporal Dispersion and ISI. Many underwater optical channel models focus primarily on average received power or path loss and neglect the temporal spreading of optical pulses caused by multiple scattering. In highly scattering waters, photons arriving via longer indirect paths introduce significant delay spread, which can result in ISI in high-speed UOWC systems. Models that do not explicitly incorporate temporal impulse response characteristics may therefore overestimate achievable data rates and underestimate error probability in bandwidth-limited scenarios.

The limitations discussed above indicate that no single physics-based underwater optical channel model is universally applicable across all water types and operating regimes. Accurate modeling of UOWC systems requires careful consideration of water clarity, scattering order, temporal dispersion, and receiver configuration, and often necessitates hybrid approaches that combine analytical modeling with Monte Carlo simulations or experimental validation.

D. Challenges of Analytical Underground Optical Propagation Models

Inaccuracy in Modeling Irregular Tunnel Geometries. Most analytical underground VLC models assume simplified tunnel geometries with planar walls and uniform cross-sections to derive NLoS channel gain. In practical mining environments, tunnels are narrow, irregular, and non-uniform, with rough and non-planar surfaces. These geometric irregularities introduce complex multipath propagation and large variability in link distance that cannot be accurately captured by idealized Lambertian reflection models. Consequently, models relying

on uniform wall reflectivity ρ may underestimate signal dispersion and fail to predict the pronounced power fluctuations observed in workface regions with rough coal surfaces.

Oversimplification of Particulate Scattering Due to Coal Dust. Analytical frameworks for underground optical channels often incorporate scattering components based on dust concentration while assuming a spatially uniform or temporally static particle distribution. In reality, underground mining environments are highly dynamic, with excavation activities continuously releasing coal dust and aerosol particles, often coupled with moisture and airflow variations. The resulting rapid fluctuations in particle size, density, and distribution can significantly increase scattering-induced attenuation and temporal dispersion. As a result, simplified scattering models may underestimate signal degradation and ISI, particularly in active excavation zones.

Stochastic Shadowing and Obstruction Effects. LoS channel models typically assume a clear and persistent optical path between the transmitter and receiver. In subterranean environments, this assumption is frequently violated due to the presence of stationary infrastructure (e.g., hydraulic supports) and mobile machinery. These obstructions introduce stochastic and often binary shadowing events, leading to abrupt signal fades and transient link outages. Deterministic analytical models are generally ill-suited to capture such non-stationary blockage behavior, resulting in optimistic reliability estimates in high-traffic mining areas.

Mechanical Misalignment and Tunnel Deformation. Physics-based underground optical models commonly assume fixed transmitter and receiver orientations. In practice, tunnel deformation, ground vibration, and unstable mounting surfaces can cause gradual or sudden rotations and tilts of LEDs and photodetectors. Such misalignment alters the irradiance and incidence angles (ϕ, ω) that govern Lambertian radiation and reception, leading to unpredictable variations in channel gain. Without adaptive or time-varying modeling, static analytical formulations may become increasingly inaccurate as infrastructure conditions evolve.

The underground optical channel presents a uniquely challenging propagation environment in which purely analytical models are often insufficient to capture the combined effects of geometric irregularity, particulate scattering, stochastic obstruction, and mechanical instability. Robust system design therefore requires hybrid modeling approaches that integrate analytical formulations with empirical measurements or data-driven techniques, such as ML, to account for the highly nonlinear and time-varying behavior of subterranean optical channels.

E. Challenges of Analytical Deep Space Optical Propagation Models

PAT sensitivity remains a fundamental challenge in DSOC due to extremely narrow optical beams and astronomical propagation distances. Even small pointing errors can cause severe signal degradation or complete link loss. Many analytical models assume ideal or stationary alignment and approximate misalignment using simplified beam displacement models. In practice, spacecraft vibrations, reaction wheel jitter, thermal deformation, and attitude control errors introduce time-varying pointing disturbances that produce beam wander and intermittent link degradation, which are difficult to capture with static analytical formulations.

Background radiation and solar interference also pose significant challenges for DSOC propagation modeling. Analytical models often emphasize geometric spreading and optical gain while assuming simplified background noise conditions. In realistic space environments, detectors are exposed to solar radiation, zodiacal light, and planetary albedo. When optical terminals operate near the Sun–spacecraft–Earth alignment or bright celestial objects, background photons can significantly increase shot noise and reduce the achievable SNR, making accurate analytical modeling difficult.

Extreme propagation distances further complicate analytical DSOC modeling. DSOC links span distances from lunar missions to interplanetary scales, where small uncertainties in pointing, optical efficiency, or receiver alignment can produce large variations in received power. Additionally, spacecraft motion and Doppler shifts introduce time-varying link geometry that static analytical models often cannot accurately capture.

These challenges indicate that analytical DSOC propagation models, while useful for first-order link analysis, rely on simplifying assumptions regarding pointing stability, background radiation, and link geometry. Accurate DSOC modeling therefore often requires hybrid approaches that integrate analytical formulations with spacecraft dynamics and environmental characterization.

F. Challenges in Empirical Propagation Modeling

Empirical propagation models complement analytical channel modeling by deriving parameterized relationships directly from experimental measurements. Rather than deriving attenuation or turbulence behavior strictly from first principles, empirical models capture the aggregate impact of complex environmental factors through statistical regression or parameterized formulas. In atmospheric optical communication, widely used examples include the Kim, Kruse, and Al-Naboulsi models, which estimate attenuation as a function of visibility and wavelength. Similar empirical approaches have also been explored in underwater, indoor, underground, and space optical communication systems. The primary advantage of empirical models lies in their ability to capture real-world propagation behavior that may be difficult to describe analytically, particularly in heterogeneous environments where scattering particles, turbulence, or medium composition vary unpredictably. By leveraging measurement campaigns and historical environmental data, empirical models can provide practical performance estimates for specific deployment scenarios. However, empirical models are typically valid only within the environmental conditions under which they were derived. When applied outside their calibration range, such models may produce inaccurate predictions due to changes in particle composition, turbulence structure, or environmental dynamics. Consequently, empirical modeling is often complemented by

analytical frameworks or, more recently, ML techniques that aim to generalize across diverse propagation conditions.

G. Challenges in ML-Based Propagation Modeling

Data scarcity and long-tail environmental events. A primary bottleneck for ML-based optical propagation modeling across atmospheric, indoor, underwater, underground, and space environments is the limited availability of high-quality, labeled datasets that capture rare but operationally critical conditions. Examples include extreme sandstorms or dense fog in atmospheric links, rapid orientation changes and shadowing events in indoor environments, highly turbid or biologically active water regimes in underwater channels, sudden coal-dust surges in underground mines, and pointing instability or solar radiation interference in space optical links. Most existing studies therefore rely on short-term measurement campaigns or synthetically generated data, which may not fully capture the statistical diversity and long-term variability of real propagation environments.

Challenge: Models trained predominantly under nominal or mildly impaired conditions may experience severe performance degradation during rare or localized environmental events, limiting reliability in safety-critical deployments.

Future direction: Federated and collaborative learning frameworks, in which geographically distributed terminals jointly train shared models without exchanging raw data, offer a promising pathway toward globally robust propagation models that reflect diverse climatic and environmental regimes.

Generalization and domain adaptation. ML-based propagation models are often highly site-specific. A model trained for a maritime atmospheric link may not generalize to desert, urban, mountainous, subterranean, or underwater environments due to differences in aerosol composition, turbulence statistics, particulate distribution, and boundary-layer dynamics.

Challenge: Domain shift, where the statistical properties of deployment data differ from those of the training dataset, leading to unpredictable performance degradation.

Future direction: Transfer learning, meta-learning, and unsupervised domain adaptation techniques can enable pre-trained propagation models to be efficiently adapted to new deployment scenarios using limited local measurements, thereby improving cross-environment robustness.

Physics-informed ML and hybrid modeling. Purely data-driven models may generate predictions that violate known physical constraints if no prior knowledge is enforced.

Challenge: Limited interpretability, lack of physical consistency, and difficulty in extrapolating beyond observed training regimes.

Future direction: Physics-informed ML approaches, including physics-informed neural networks (PINNs) and hybrid analytical–learning frameworks, can embed governing relationships (e.g., Beer–Lambert attenuation, turbulence-induced variance models, or radiative transfer formulations) into the training objective, improving interpretability and out-of-distribution robustness.

Hardware constraints and real-time inference. Although ML models can provide fast inference once trained, complex deep architectures may impose significant computational and energy overheads for low-power IoT devices, UAV platforms, or satellite-based optical terminals.

Challenge: Balancing model complexity with strict latency, memory, and power constraints, particularly for real-time tasks such as PAT or adaptive beam control.

Future direction: Model compression, pruning, quantization, TinyML techniques, and hardware-aware neural architecture design can enable efficient deployment of learning-based propagation models on edge devices and FPGA/ASIC-based optical transceivers.

Standardized benchmarking and open datasets. Unlike mature ML domains such as computer vision, OWC lacks standardized propagation datasets and benchmarking protocols spanning multiple environments and operating regimes.

Challenge: Heterogeneous experimental setups, inconsistent performance metrics, and limited reproducibility hinder fair comparison across analytical, empirical, and ML-based studies.

Future direction: Development of open-access, multi-regime optical wireless datasets with synchronized environmental measurements, received-signal statistics, and turbulence or scattering indicators would significantly accelerate reproducible and comparable ML research across atmospheric, underwater, and underground systems.

These challenges highlight that, while ML-based models offer flexibility and adaptability, they cannot fully replace physics-based insight or empirical validation. A unified future research direction lies in hybrid frameworks that integrate analytical interpretability, empirical calibration, and data-driven adaptability to enable robust optical propagation modeling across diverse environments.

XII. Conclusion

This survey presented a unified review of optical wireless propagation models across atmospheric, space, underwater, indoor, and underground environments. By comparing analytical, empirical, and ML-based approaches, the paper highlighted the key physical mechanisms such as turbulence, scattering, absorption, multipath, and pointing errors that shape channel behavior in each environment. Analytical models provide essential physical insight, empirical models capture real world variability, and ML offers adaptability where traditional models are limited. The comparison shows that propagation modeling must remain specific to each environment, since no single framework can describe all operational conditions. Looking forward, combining physics based models with data driven methods, enabling real time adaptive channel estimation, and developing unified modeling frameworks for hybrid RF and optical systems are important opportunities for advancing next generation OWC.

References

[1] S. A. Al-Gailani, M. F. M. Salleh, A. A. Salem, R. Q. Shaddad, U. U. Sheikh, N. A. Algeelani, and T. A. Almohamad, "A survey of free space optics (fso) communication systems, links, and networks," *IEEE Access*, vol. 9, pp. 7353–7373, 2020.

[2] H. E. Nistazakis, T. A. Tsiftsis, and G. S. Tombras, "Performance analysis of free-space optical communication systems over atmospheric turbulence channels," *IET communications*, vol. 3, no. 8, pp. 1402–1409, 2009.

[3] M. Goncalves Teixeira, J. R. Molina, and V. N. Soares, "Review on free-space optical communications for delay and disruption tolerant networks," *Electronics*, vol. 10, no. 13, p. 1607, 2021.

[4] H. Kaushal and G. Kaddoum, "Free space optical communication: challenges and mitigation techniques," *arXiv preprint arXiv:1506.04836*, 2015.

[5] P. W. Kruse, L. D. McGlauchlin, and R. B. McQuistan, "Elements of infrared technology: Generation, transmission and detection," *New York: Wiley*, 1962.

[6] M. Al Naboulsi, H. Sizun, and F. d. r. de Fornel, "Fog attenuation prediction for optical and infrared waves," *Optical Engineering*, vol. 43, no. 2, pp. 319–329, 2004.

[7] M. A. Khalighi and M. Uysal, "Survey on free space optical communication: A communication theory perspective," *IEEE communications surveys & tutorials*, vol. 16, no. 4, pp. 2231–2258, 2014.

[8] S. Sahu and P. K. Sahu, "Review on performance enhancement techniques in fso systems," *IETE Technical Review*, pp. 1–23, 2025.

[9] T. K. Sarkar, Z. Ji, K. Kim, A. Medouri, and M. Salazar-Palma, "A survey of various propagation models for mobile communication," *IEEE Antennas and propagation Magazine*, vol. 45, no. 3, pp. 51–82, 2003.

[10] Z. Ghassemlooy, W. Popoola, and S. Rajbhandari, *Optical wireless communications: system and channel modelling with Matlab®*. CRC press, 2019.

[11] A. Al-Kinani, C.-X. Wang, L. Zhou, and W. Zhang, "Optical wireless communication channel measurements and models," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 3, pp. 1939–1962, 2018.

[12] F. Miramirkhani and M. Uysal, "Channel modelling for indoor visible light communications," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 378, no. 2169, p. 20190187, 2020.

[13] D. Anguita, D. Brizzolara, G. Parodi, G. Merrett, and Y. Tan, "Prospects and problems of optical diffuse wireless communication for underwater wireless sensor networks (uwsns)," *Wireless Sensor Networks: Application-Centric Design*, pp. 275–300, 2010.

[14] R. A. Moffione, "Evolution and revolution in measuring ocean optical properties," *Oceanography*, vol. 14, no. 3, pp. 9–14, 2001.

[15] L. J. Johnson, "The underwater optical channel," *no. November*, pp. 1–18, 2012.

[16] P. Sajmath, R. V. Ravi, and K. A. Majeed, "Underwater wireless optical communication systems: A survey," in *2020 7th international conference on smart structures and systems (ICSSS)*, pp. 1–7, IEEE, 2020.

[17] S. Arnon, "Underwater optical wireless communication network," *Optical Engineering*, vol. 49, no. 1, pp. 015001–015001, 2010.

[18] H. Kaushal and G. Kaddoum, "Underwater optical wireless communication," *IEEE access*, vol. 4, pp. 1518–1547, 2016.

[19] N. Saeed, M.-S. Alouini, and T. Y. Al-Naffouri, "Toward the internet of underground things: A systematic survey," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 4, pp. 3443–3466, 2019.

[20] S. M. Riurean, M. Leba, and A. C. Ionica, *Application of visible light wireless communication in underground mine*. Springer, 2021.

[21] S. Karmous, N. Adem, M. Atiquzzaman, and S. Samarakoon, "How can optical communications shape the future of deep space communications? a survey," *IEEE Communications Surveys & Tutorials*, vol. 27, no. 1, pp. 725–747, 2024.

[22] W. Wu, M. Chen, Z. Zhang, X. Liu, and Y. Dong, "Overview of deep space laser communication," *Science China Information Sciences*, vol. 61, no. 4, p. 040301, 2018.

[23] H. Ivanov, S. Mejri, A. Di Mira, K.-J. Schulz, and C. Heese, "Review of deep space optical communications," *International Journal of Satellite Communications and Networking*, vol. 43, no. 3, pp. 193–209, 2025.

[24] H. Kaushal and G. Kaddoum, "Optical communication in space: Challenges and mitigation techniques," *IEEE communications surveys & tutorials*, vol. 19, no. 1, pp. 57–96, 2016.

[25] R. Li, B. Lin, Y. Liu, M. Dong, and S. Zhao, "A survey on laser space network: terminals, links, and architectures," *IEEE Access*, vol. 10, pp. 34815–34834, 2022.

[26] I. I. Kim, B. McArthur, and E. J. Korevaar, "Comparison of laser beam propagation at 785 nm and 1550 nm in fog and haze for optical wireless communications," in *Optical wireless communications III*, vol. 4214, pp. 26–37, Spie, 2001.

[27] A. Patil, S. Iyer, and R. J. Pandya, "A survey of machine learning algorithms for 6g wireless networks," *arXiv preprint arXiv:2203.08429*, 2022.

[28] M. I. Jordan and T. M. Mitchell, "Machine learning: Trends, perspectives, and prospects," *Science*, vol. 349, no. 6245, pp. 255–260, 2015.

[29] T. Hastie, R. Tibshirani, J. Friedman, *et al.*, "The elements of statistical learning," 2009.

[30] A. Lionis, K. Peppas, H. E. Nistazakis, A. Tsigopoulos, K. Cohn, and A. Zagouras, "Using machine learning algorithms for accurate received optical power prediction of an fso link over a maritime environment," in *Photonics*, vol. 8, p. 212, MDPI, 2021.

[31] Z. M. C. Lima, R. Costa, Y. Martínez-Camejo, A. I. L. de Araujo, and G. de Freitas Gilmarães, "Tuning the parameters of a free-space optical channel using machine learning," *Applied Optics*, vol. 63, no. 11, pp. 2876–2881, 2024.

[32] S. Bloom, E. Korevaar, J. Schuster, and H. Willebrand, "Understanding the performance of free-space optics," *Journal of optical Networking*, vol. 2, no. 6, pp. 178–200, 2003.

[33] D. Killinger, "Free space optics for laser communication through the air," *Optics and photonics news*, vol. 13, no. 10, pp. 36–42, 2002.

[34] A. Sevincer, A. Bhattarai, M. Bilgi, M. Yuksel, and N. Pala, "Lightnets: Smart lighting and mobile optical wireless networks—a survey," *IEEE Communications Surveys & Tutorials*, vol. 15, no. 4, pp. 1620–1641, 2013.

[35] A. A. Farid and S. Hranilovic, "Outage capacity optimization for free-space optical links with pointing errors," *Journal of Lightwave technology*, vol. 25, no. 7, pp. 1702–1710, 2007.

[36] N. R. Barbeau, "Power deposited by a gaussian beam on a decentered circular aperture," *Applied optics*, vol. 34, no. 28, pp. 6443–6445, 1995.

[37] H. Willebrand and B. S. Ghuman, *Free space optics: enabling optical connectivity in today's networks*. SAMS publishing, 2002.

[38] E. J. Korevaar, "Optical wireless communications v," *Optical Wireless Communications V*, vol. 4873, 2002.

[39] L. C. Andrews and R. L. Phillips, *Laser beam propagation through random media*. Bellingham, WA: SPIE Press, second ed., 2005.

[40] Y. Hu, H. Bian, R. Wang, Z. Wen, and S. Hu, "Statistical analysis of positioning errors in long-endurance dual-axis rotary modulation inertial navigation system," *Electronics*, vol. 13, no. 23, p. 4671, 2024.

[41] M. Nakagami, "The m-distribution—a general formula of intensity distribution of rapid fading," in *Statistical methods in radio wave propagation*, pp. 3–36, Elsevier, 1960.

[42] M. S. U. Chowdhury and M. Yuksel, "Optimum tiling of free-space optical transceivers for in-band full-duplex communication," *IEEE Transactions on Communications*, vol. 74, pp. 1923–1939, 2025.

[43] T. Kaur, J. Singh, and A. Sharma, "Simulative analysis of rayleigh and rician fading channel model and its mitigation," in *2017 8th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, pp. 1–6, IEEE, 2017.

[44] E. Zedini, Y. Ata, A. Kammoun, and M.-S. Alouini, "A novel approach to approximating generalized pointing errors modeled by beckmann distribution in fso communication systems," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 727–741, 2024.

[45] B. Zhu, Z. Zeng, J. Cheng, and N. C. Beaulieu, "On the distribution function of the generalized beckmann random variable and its applications in communications," *IEEE Transactions on Communications*, vol. 66, no. 5, pp. 2235–2250, 2018.

[46] A. F. Haq and M. Yuksel, "Weather-limited in-band full-duplex transceiver model for free-space optical communication," *Optical Engineering*, vol. 59, no. 5, pp. 056113–056113, 2020.

[47] A. S. Haq and M. Yuksel, "Multi-element full-duplex fso transceiver design using optimum tiling," *Optics Communications*, vol. 501, p. 127377, 2021.

[48] M. S. U. Chowdhury and M. Yuksel, "In-band full-duplex free-space optical transceiver design for flying platforms," in *MILCOM 2023-2023 IEEE Military Communications Conference (MILCOM)*, pp. 438–443, IEEE, 2023.

[49] M. Ijaz, Z. Ghassemlooy, J. Pesek, O. Fiser, H. Le Minh, and E. Bentley, "Modeling of fog and smoke attenuation in free space optical communications link under controlled laboratory conditions," *Journal of Lightwave Technology*, vol. 31, no. 11, pp. 1720–1726, 2013.

[50] M. A. Esmail, H. Fathallah, and M.-S. Alouini, "Outdoor fso communications under fog: attenuation modeling and performance evaluation," *IEEE photonics journal*, vol. 8, no. 4, pp. 1–22, 2016.

[51] B. Epple, "Simplified channel model for simulation of free-space optical communications," *Journal of Optical Communications and networking*, vol. 2, no. 5, pp. 293–304, 2010.

[52] A. Lionis, A. Sklavounos, A. Stassinakis, K. Cohn, A. Tsigopoulos, K. Peppas, K. Aidinis, and H. Nistazakis, "Experimental machine learning approach for optical turbulence and fso outage performance modeling," *Electronics*, vol. 12, no. 3, p. 506, 2023.

[53] N. Maharjan and B. W. Kim, "Machine learning-based beam pointing error reduction for satellite–ground fso links," *Electronics*, vol. 13, no. 17, p. 3466, 2024.

[54] L. J. S. Kumar, P. Krishnan, B. Shreya, and S. MS, "Performance enhancement of fso communication system using machine learning for 5g/6g and iot applications," *Optik*, vol. 252, p. 168430, 2022.

[55] M. A. Amirabadi, M. H. Kahaei, and S. A. Nezamalhosseini, "Deep learning based detection technique for fso communication systems," *Physical Communication*, vol. 43, p. 101229, 2020.

[56] M. A. Esmail, W. S. Saif, A. M. Ragheb, and S. A. Alshebeili, "Free space optic channel monitoring using machine learning," *Optics Express*, vol. 29, no. 7, pp. 10967–10981, 2021.

[57] Z. Ghassemlooy, W. Popoola, and S. Rajbhandari, *Optical wireless communications: System and channel modelling with MATLAB*. CRC Press, 2013.

[58] H. Farahneh, C. Mekhiel, A. Khalifeh, W. Farjow, and X. Fernando, "Shadowing effects on visible light communication channels," in *2016 IEEE Canadian conference on electrical and computer engineering (CCECE)*, pp. 1–5, IEEE, 2016.

[59] J. M. Kahn and J. R. Barry, "Wireless infrared communications," *Proceedings of the IEEE*, vol. 85, no. 2, pp. 265–298, 2002.

[60] C. Singh, J. John, Y. N. Singh, and K. K. Tripathi, "A review of indoor optical wireless systems," *IETE Technical Review*, vol. 19, no. 1-2, pp. 3–17, 2002.

[61] K. Lee, H. Park, and J. R. Barry, "Indoor channel characteristics for visible light communications," *IEEE Communications Letters*, vol. 15, no. 2, pp. 217–219, 2011.

[62] P. Chvojka, S. Zvanovec, P. A. Haigh, and Z. Ghassemlooy, "Channel characteristics of visible light communications within dynamic indoor environment," *Journal of Lightwave Technology*, vol. 33, no. 9, pp. 1719–1725, 2015.

[63] W. Yang, L. Qin, X. Hu, and D. Zhao, "Indoor visible-light 3d positioning system based on gru neural network," in *Photonics*, vol. 10, p. 633, MDPI, 2023.

[64] M. E. Hosney, H. A. I. Selmy, and K. M. F. Elsayed, "Co-channel interference management in visible light communication," in *2020 29th Wireless and Optical Communications Conference (WOCC)*, pp. 1–4, 2020.

[65] A. Kaba, S. Sahuguede, and A. Julien-Vergonjanne, "Channel modeling of an optical wireless body sensor network for walk monitoring of elderly," *Sensors*, vol. 21, no. 9, 2021.

[66] H. Li, J. Wang, X. Zhang, and R. Wu, "Indoor visible light positioning combined with ellipse-based aco-ofdm," *IET Communications*, vol. 12, no. 17, pp. 2181–2187, 2018.

[67] L. Zeng, D. C. O'Brien, H. Le Minh, G. E. Faulkner, K. Lee, D. Jung, Y. Oh, and E. T. Won, "High data rate multiple input multiple output (mimo) optical wireless communications using white led lighting," *IEEE Journal on Selected Areas in Communications*, vol. 27, no. 9, pp. 1654–1662, 2009.

[68] M. S. Almothafar, A. A. Al-Joda, and H. H. Abbas, "A review of indoor li-fi receiver challenges: Mathematical models and testing approaches," *Al-Furat Journal of Innovations in Electronics and Computer Engineering*, vol. 4, no. 1, pp. 236–256, 2025.

[69] Z. Wang, Z. Liang, X. Li, and H. Li, "Indoor visible light positioning based on improved particle swarm optimization method with min-max algorithm," *IEEE Access*, vol. 10, pp. 130068–130077, 2022.

[70] N.-T. Nguyen, A. Suebsomran, K. Sripimanwat, and N.-H. Nguyen, "Design and simulation of a novel indoor mobile robot localization method using a light-emitting diode positioning system," *Transactions of the Institute of Measurement and Control*, vol. 38, no. 3, pp. 305–314, 2016.

[71] C. Chen, D. A. Basnayaka, X. Wu, and H. Haas, "Efficient analytical calculation of non-line-of-sight channel impulse response in visible light communications," *Journal of Lightwave Technology*, vol. 36, no. 9, pp. 1666–1682, 2017.

[72] S. Sinha and A. Chockalingam, "Otfs modulation in dual-led indoor visible light communication systems," in *2021 IEEE 94th Vehicular Technology Conference (VTC2021-Fall)*, pp. 1–7, IEEE, 2021.

[73] Y. Qiu, H.-H. Chen, and W.-X. Meng, "Channel modeling for visible light communications—a survey," *Wireless Communications and Mobile Computing*, vol. 16, no. 14, pp. 2016–2034, 2016.

[74] P. Tang, Y. Yin, Y. Tong, S. Liu, L. Li, T. Jiang, Q. Wang, and M. Chen, "Channel characterization and modeling for vlc-ioe applications in 6g: A survey," *IEEE Internet of Things Journal*, 2024.

[75] C. Chen, *Advanced Modulation and Detection Techniques for Indoor Visible Light Communication Systems*. PhD thesis, Nanyang Technological University, 2017.

[76] H. Lu, Z. Su, and B. Yuan, "Snr and optical power distribution in an indoor visible light communication system," in *2014 7th International Congress on Image and Signal Processing*, pp. 1063–1067, IEEE, 2014.

[77] M. M. Ali and K. F. Hussein, "Intersymbol interference in indoor visible light communication systems employing led sources of large area," *The Egyptian International Journal of Engineering Sciences and Technology*, vol. 38, no. 2, pp. 60–74, 2022.

[78] T. Komine and M. Nakagawa, "Fundamental analysis for visible-light communication system using led lights," *IEEE Transactions on Consumer Electronics*, vol. 50, no. 1, pp. 100–107, 2004.

[79] P. Pešek, S. Zvanovec, P. Chvojka, M. R. Bhatnagar, Z. Ghassemlooy, and P. Saxena, "Mobile user connectivity in relay-assisted visible light communications," *Sensors*, vol. 18, no. 4, p. 1125, 2018.

[80] J. R. Barry, J. M. Kahn, W. J. Krause, E. A. Lee, and D. G. Messerschmitt, "Simulation of multipath impulse response for indoor wireless optical channels," *IEEE journal on selected areas in communications*, vol. 11, no. 3, pp. 367–379, 1993.

[81] K. Lee, H. Park, and J. R. Barry, "Indoor channel characteristics for visible light communications," *IEEE communications letters*, vol. 15, no. 2, pp. 217–219, 2011.

[82] S. Rodríguez Pérez, R. Pérez Jiménez, F. J. López Hernández, O. B. González Hernández, and A. J. Ayala Alfonso, "Reflection model for calculation of the impulse response on ir-wireless indoor channels using ray-tracing algorithm," *Microwave and Optical Technology Letters*, vol. 32, no. 4, pp. 296–300, 2002.

[83] M. Uysal, F. Miramirkhani, O. Narmanlioglu, T. Baykas, and E. Panayirci, "Ieee 802.15. 7r1 reference channel models for visible light communications," *IEEE Communications Magazine*, vol. 55, no. 1, pp. 212–217, 2017.

[84] Y. Tong, P. Tang, J. Zhang, S. Liu, Y. Yin, B. Liu, and L. Xia, "Channel characteristics and link adaption for visible light communication in an industrial scenario," *Sensors*, vol. 23, no. 7, p. 3442, 2023.

[85] F. Miramirkhani, "A path loss model for link budget analysis of indoor visible light communications," *Electrica*, vol. 21, no. 2, 2021.

[86] Y. S. Eroğlu, Y. Yapıcı, and I. Güvenç, "Impact of random receiver orientation on visible light communications channel," *IEEE Transactions on Communications*, vol. 67, no. 2, pp. 1313–1325, 2018.

[87] X.-T. Fu, R.-R. Lu, and J.-Y. Wang, "Realistic performance analysis for visible light communication with random receivers," *J. Opt. Soc. Am. A*, vol. 38, pp. 654–662, May 2021.

[88] M. M. Céspedes, B. G. Guzmán, and V. P. G. Jiménez, "Lights and shadows: A comprehensive survey on cooperative and precoding schemes to overcome los blockage and interference in indoor vlc," *Sensors*, vol. 21, no. 3, p. 861, 2021.

[89] D. Luan and J. Thompson, "Robust channel estimation for optical wireless communications using neural network," 2025.

[90] D. Luan and J. Thompson, "Low complexity channel estimation with neural network solutions," in *WSA 2021; 25th International ITG Workshop on Smart Antennas*, pp. 1–6, VDE, 2021.

[91] Z. Zeng, S. Fu, H. Zhang, Y. Dong, and J. Cheng, "A survey of underwater optical wireless communications," *IEEE communications surveys & tutorials*, vol. 19, no. 1, pp. 204–238, 2016.

[92] H. Ochi, Y. Watanabe, and T. Shimura, "Experiments of underwater acoustic communication using 16-qam," in *8th International Congress on Acoustics*, 2004.

[93] I. F. Akyildiz, D. Pompili, and T. Melodia, "Underwater acoustic sensor networks: research challenges," *Ad hoc networks*, vol. 3, no. 3, pp. 257–279, 2005.

[94] T. J. Hayward and T. Yang, "Underwater acoustic communication channel capacity: A simulation study," in *AIP conference proceedings*, vol. 728, pp. 114–121, American Institute of Physics, 2004.

[95] M. A. Chancey, "Short range underwater optical communication links," Master's thesis, North Carolina State University, 2005.

[96] S. Jaruwatanadilok, "Underwater wireless optical communication channel modeling and performance evaluation using vector radiative transfer theory," *IEEE Journal on selected areas in communications*, vol. 26, no. 9, pp. 1620–1627, 2008.

[97] S. S. Reza and M. Yuksel, "UAV trajectory optimization for air-to-water optical wireless channels," in *2024 33rd International Conference on Computer Communications and Networks (ICCCN)*, pp. 1–9, IEEE, 2024.

[98] J. W. Giles and I. N. Bankman, "Underwater optical communications systems. part 2: basic design considerations," in *MILCOM 2005-2005 IEEE Military Communications Conference*, pp. 1700–1705, IEEE, 2005.

[99] M. T. Nguyen, V. V. Mai, and C. T. Nguyen, "Sinr performance analysis of 3-d underwater optical wireless communication networks," in *2019 26th international conference on telecommunications (ICT)*, pp. 41–45, IEEE, 2019.

[100] H. Lu, M. Jiang, and J. Cheng, "Deep learning aided robust joint channel classification, channel estimation, and signal detection for underwater optical communication," *IEEE Transactions on Communications*, vol. 69, no. 4, pp. 2290–2303, 2021.

[101] I. Romdhane and G. Kaddoum, "A reinforcement-learning-based beam adaptation for underwater optical wireless communications," *IEEE Internet of Things Journal*, vol. 9, no. 20, pp. 20270–20281, 2022.

[102] C. Wang, Z. Wang, W. Sun, and D. R. Fuhrmann, "Reinforcement learning-based adaptive transmission in time-varying underwater acoustic channels," *IEEE access*, vol. 6, pp. 2541–2558, 2017.

[103] Q. Fu and A. Song, "Adaptive modulation for underwater acoustic communications based on reinforcement learning," in *OCEANS 2018 MTS/IEEE Charleston*, pp. 1–8, IEEE, 2018.

[104] H. Wang, Y. Li, and J. Qian, "Self-adaptive resource allocation in underwater acoustic interference channel: A reinforcement learning approach," *IEEE Internet of Things Journal*, vol. 7, no. 4, pp. 2816–2827, 2019.

[105] R. Wang, A. Yadav, E. A. Makled, O. A. Dobre, R. Zhao, and P. K. Varshney, "Optimal power allocation for full-duplex underwater relay networks with energy harvesting: A reinforcement learning approach," *IEEE wireless communications letters*, vol. 9, no. 2, pp. 223–227, 2019.

[106] P. Palacios Játiva, C. A. Azurdia-Meza, I. Sánchez, D. Zabala-Blanco, A. Dehghan Firoozabadi, I. Soto, and F. Seguel, "An enhanced vlc channel model for underground mining environments considering a 3d dust particle distribution model," *Sensors*, vol. 22, no. 7, p. 2483, 2022.

[107] O. Stoicuta, S. Riurean, S. Burian, M. Leba, and A. Ionica, "Application of optical communication for an enhanced health and safety system in underground mine," *Sensors*, vol. 23, no. 2, p. 692, 2023.

[108] Y. Zhai and S. Zhang, "Visible light communication channel models and simulation of coal workface energy coupling," *Mathematical problems in engineering*, vol. 2015, no. 1, p. 271352, 2015.

[109] P. P. Játiva, C. A. Azurdia-Meza, M. R. Cañizares, I. Sánchez, and D. Iturralde, "On the performance of visible light communications in underground mines," in *2020 IEEE Latin-American Conference on Communications (LATINCOM)*, pp. 1–6, IEEE, 2020.

[110] S. Riurean, O. Stoicuta, M. Leba, A. Ionica, and Á. Rocha, "Underground channel model for visible light wireless communication based on neural networks," in *World Conference on Information Systems and Technologies*, pp. 293–305, Springer, 2020.

[111] Z. Sodnik, C. Heese, P.-D. Arapoglou, K.-J. Schulz, I. Zayer, R. Daddato, and S. Kraft, "Deep-space optical communication system (docs) for esa's space weather mission to lagrange orbit l5," in *2017 IEEE international conference on space optical systems and applications (ICSOS)*, pp. 28–33, IEEE, 2017.

[112] A. Könsgen and A. Förster, "Current state and future challenges in deep space communication: A survey," *it-Information Technology*, vol. 63, no. 4, pp. 219–234, 2021.

[113] G. Xu and Z. Song, "Effects of solar scintillation on deep space communications: Challenges and prediction techniques," *IEEE Wireless Communications*, vol. 26, no. 2, pp. 10–16, 2019.

[114] M. R. Patel and O. Beik, *Spacecraft power systems*. CRC press, 2023.

[115] P. Chetty, "Electrical power system for low-earth-orbit spacecraft applications," *Journal of Propulsion and Power*, vol. 6, no. 1, pp. 63–68, 1990.

[116] C.-Y. Yeh and E. W. Knightly, "Eavesdropping in massive mimo: New vulnerabilities and countermeasures," *IEEE Transactions on Wireless Communications*, vol. 20, no. 10, pp. 6536–6550, 2021.

[117] J. A. Mendenhall, L. M. Candell, P. I. Hopman, G. Zogbi, D. M. Boroson, D. O. Caplan, C. J. Digenis, D. R. Hearn, and R. C. Shoup, "Design of an optical photon counting array receiver system for deep-space communications," *Proceedings of the IEEE*, vol. 95, no. 10, pp. 2059–2069, 2007.

[118] N. K. Lyras, A. D. Panagopoulos, and P.-D. Arapoglou, "Deep-space optical communication link engineering: Sensitivity analysis," *IEEE Aerospace and Electronic Systems Magazine*, vol. 34, no. 11, pp. 8–19, 2019.

[119] S. L. Booth and R. M. Manning, "Optical communication link assessment utilizing a modulated retro-reflector on mars," tech. rep., 2018.

[120] S. Karmous and N. Adem, "Connecting the universe: Challenges, mitigation, advances, and link engineering," *arXiv preprint arXiv:2302.09518*, 2023.

[121] S. M. Flatté and J. S. Gerber, "Irradiance-variance behavior by numerical simulation for plane-wave and spherical-wave optical propagation through strong turbulence," *Journal of the Optical Society of America A*, vol. 17, no. 6, pp. 1092–1097, 2000.

[122] J. A. Shaw, "Radiometry and the friis transmission equation," *American journal of physics*, vol. 81, no. 1, pp. 33–37, 2013.

[123] J. Alda, "Laser and gaussian beam propagation and transformation," *Encyclopedia of optical engineering*, vol. 999, p. 1013, 2003.

[124] C. F. Bohren and E. Clothiaux, *Fundamentals of atmospheric radiation*. Wiley Online Library, 2005.

[125] B. W. Carroll and D. A. Ostlie, *An introduction to modern astrophysics*. Cambridge University Press, 2017.

[126] L. C. Andrews and M. K. Beason, *Laser beam propagation in random media: new and advanced topics*. 2023.

[127] H. Hemmati, "Near-earth laser communications," in *Near-Earth Laser Communications, Second Edition*, pp. 1–40, CRC press, 2020.

[128] B. Moision, J. Wu, and S. Shambayati, "An optical communications link design tool for long-term mission planning for deep-space missions," in *2012 IEEE Aerospace Conference*, pp. 1–12, IEEE, 2012.

[129] H. Hemmati, *Deep space optical communications*. John Wiley & Sons, 2006.

[130] M. P. Bart, N. J. Savino, P. Regmi, L. Cohen, H. Safavi, H. C. Shaw, S. Lohani, T. A. Searles, B. T. Kirby, and H. Lee, "Deep learning for enhanced free-space optical communications," *Machine Learning: Science and Technology*, vol. 4, no. 4, p. 045046, 2023.

[131] S. Lohani, E. M. Knutson, and R. T. Glasser, "Generative machine learning for robust free-space communication," *Communications Physics*, vol. 3, no. 1, p. 177, 2020.

[132] C. He and S. Collins, "Signal demodulation using a radial basis function neural network (rbfnn) in a silicon photomultiplier-based visible light communication system," *IEEE Photonics Journal*, vol. 14, no. 4, pp. 1–14, 2022.

[133] A. Ahmed and M. B. Rahaim, "gr-owc: An open source gnuradio-based toolkit for optical wireless communications," in *2021 17th International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob)*, pp. 48–53, 2021.

[134] R. T. Wilson, "Py6s: A python interface to the 6s radiative transfer model," *Computers & Geosciences*, vol. 51, pp. 166–171, 2013.